



# **ACADEMIC REGULATIONS & SYLLABUS 2025-26**

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**Faculty of Computer Science & Applications**  
**Smt. Chandaben Mohanbhai Patel Institute of**  
**Computer Applications**  
**Master of Science in Information Technology**  
**M.Sc.(IT) Programme**  
**(as per CHOICE BASED CREDIT SYSTEM- (CBCS))**

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**Charotar University of Science and Technology (CHARUSAT)**  
**CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District:**  
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# ACADEMIC RULES

To ensure uniform system of education, duration of post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

## 1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) at Master's levels. Each semester will be at least 90 working days duration. Every enrolled student will be required to take a specified load of course work in the chosen course of specialization and also complete a project/dissertation if any.

## 2. Duration of Programme

| Postgraduate Programme | Master of Science in Information Technology<br>M.Sc.(IT) |
|------------------------|----------------------------------------------------------|
| Minimum                | 4 Semesters (2 Academic years)                           |
| Maximum                | 6 Semesters (3 Academic years)                           |

### **3. Eligibility & Mode of Admissions**

Eligibility of a candidate and mode of admission to the programme will be according to the regulations for admission committee decided by Government of Gujarat from time to time.

### **4. Programme structure and Credits**

A student admitted to a program should study the course and earn credits specified in the course structure.

### **5. Attendance**

5.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

5.2 Student attendance in a course should be 80%.

## **6. Course Evaluation**

6.1 The performance of every student in each course will be evaluated as follows:

- 6.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 50% of the marks for the course.
- 6.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 50% of the marks for the course.

6.2 University Examination

- 6.2.1 The final examination by the University for 50% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.
- 6.2.2 In order to earn the credit in a course a student has to obtain grade other than FF.
- 6.2.3 Performance at Internal & University Examination will be done on the relative grading system.

## **7. Grading**

The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are defined as follows:

| <b>Grading Scheme Range of Marks (%)</b> | 96.0-100                  | 86.0-95.9                | 76.0-85.9               | 66.0-75.9           | 56.0-65.9                   | 46.0 – 55.9           | 36.0 – 45.9        | Below 36.0         | Absent                |
|------------------------------------------|---------------------------|--------------------------|-------------------------|---------------------|-----------------------------|-----------------------|--------------------|--------------------|-----------------------|
| <b>Letter Grade</b>                      | <b>O</b><br>(Outstanding) | <b>A+</b><br>(Excellent) | <b>A</b><br>(Very Good) | <b>B+</b><br>(Good) | <b>B</b><br>(Above Average) | <b>C</b><br>(Average) | <b>P</b><br>(Pass) | <b>F</b><br>(Fail) | <b>Ab</b><br>(Absent) |
| <b>Grade Point</b>                       | <b>10</b>                 | <b>9</b>                 | <b>8</b>                | <b>7</b>            | <b>6</b>                    | <b>5</b>              | <b>4</b>           | <b>0</b>           | <b>0</b>              |

$SGPA = \sum C_i G_i / \sum C_i$  where  $C_i$  is the number of credits of course  $i$   $G_i$  is the Grade Point for the course  $i$  and  $i = 1$  to  $n$ ,  $n$  = number of courses in the semester.

$CGPA = \sum C_i G_i / \sum C_i$  where  $C_i$  is the number of credits of course  $i$   $G_i$  is the Grade Point for the course  $i$  and  $i = 1$  to  $n$ ,  $n$  = number of courses of all semesters up to which CGPA is computed.

## 8. Detention Rule

A student will be promoted to next year only if he/she has cleared all the courses of the year he/she is studying in.

Awards of Degree: Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

8.1.1 He should have earned at least minimum required credits as prescribed in course structure.

8.1.2 He should have cleared all evaluation components in every course.

## 9. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

|                      |                                                                |
|----------------------|----------------------------------------------------------------|
| <b>Distinction:</b>  | <b>CGPA <math>\geq 7.0</math> &amp; <math>\leq 10.0</math></b> |
| <b>First class:</b>  | <b>CGPA <math>\geq 6.0</math> &amp; <math>&lt; 7.0</math></b>  |
| <b>Second Class:</b> | <b>CGPA <math>\geq 5.0</math> &amp; <math>&lt; 6.0</math></b>  |
| <b>Pass Class:</b>   | <b>CGPA <math>\geq 3.60</math> &amp; <math>&lt; 5.0</math></b> |

## 10. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

## **CHOICE BASED CREDIT SYSTEM**

The choice based credit system provides flexibility in designing curriculum and assigning credits based on the course content and hour of teaching. The choice based credit system provides an opportunity for the students to choose courses from the prescribed courses comprising core, elective and open elective courses. The CBCS provides a cafeteria type approach in which the students can take courses of their choice and adopt an interdisciplinary approach to learning. The courses shall be evaluated on the grading system, which is considered to be better than the conventional marks system.

### **CBCS – Conceptual Definitions / Key Terms (Terminologies)**

Types of Courses: The Programme Structure consist of 3 types of courses:

Foundation Courses, Core Courses and Elective Courses.

#### **Foundation Course**

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

#### **Core Courses**

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

##### **A. University Core Courses(UC):**

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme.

##### **B. Programme Core Courses(PC):**

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

## **Elective Courses**

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialized or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates

proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

### **A. University Elective Courses(UE):**

The pool of elective courses offered across all faculties / programmes.

### **B. Institute Elective Course (IE)**

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization.

### **C. Programme Elective Courses(PE):**

The programme specific pool of elective courses offered by respective programme.

## **Vision, Mission, PEOs, POs and PSOs**

### **Vision**

To become a leading institution in the field of computer applications and contribute in national efforts of computerizing public systems.

### **Mission**

To produce competent computer professionals with the ability to face future challenges.

### **PROGRAM EDUCATIONAL OBJECTIVES (PEOs)**

The graduates will:

**PEO1:** Be able to understand the requirement of computing problem and implement an effective solution.

**PEO2:** Be able to successfully take up various available career options.

**PEO3:** Be able to continuously learn in their preferred domains.

**PEO4:** Be able to acquit themselves in ethical and professional manner while demonstrating effective skills.

### **PROGRAM OUTCOMES (POs)**

The Graduate of Computer Science and Applications will be able to:

**PO1. Computational Knowledge:** Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.

**PO2. Problem Analysis:** Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.

**PO3. Design / Development of Solutions:** Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand and propose integrated solutions using emerging technologies

**PO4. Conduct Investigations of Complex Computing Problems:** Ability to devise and conduct experiments, interpret data and provide well informed conclusions.

**PO5. Modern Tool Usage:** Ability to select modern computing tools, skills and techniques necessary for innovative software solutions

**PO6. Professional Ethics:** Ability to apply and commit professional ethics and cyber regulations in a global economic environment.

**PO7. Life-long Learning:** Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.

**PO8. Project Management and Finance:** Ability to understand, management and computing principles with computing knowledge to manage projects in multidisciplinary environments.

**PO9. Communication Efficacy:** Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations.

**PO10. Societal & Environmental Concern:** Ability to recognize economical, environmental, social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.

**PO11. Individual & Team Work:** Ability to work as a member or leader in diverse teams in multidisciplinary environment.

**PO12. Innovation and Entrepreneurship:** Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.

## **PROGRAM SPECIFIC OUTCOMES (PSOs)**

At the end of the programme, the student should be able to

**PSO1:** Analyse, design and develop effective software applications for solving contemporary challenges.

**PSO2:** Harness the necessary skills with spirit of research and entrepreneurship.

**Total Credits of Programme (4 Semesters): 108**

**TEACHING SCHEME AND  
EXAMINATION SCHEME  
FOR  
M.Sc. (IT) PROGRAMME  
EFFECTIVE FROM  
ACADEMIC YEAR 2025-26**

**Teaching & Examination Scheme**

**Master of Science in Information Technology (M.Sc. (IT)) Programme**

(Choice Based Credit System)

**Effective from Year 2025 – 26**

**Semester – I**

| Course Code                        | Course Title                              | Teaching Scheme |           |       |        | Examination Scheme |          |           |          |       |
|------------------------------------|-------------------------------------------|-----------------|-----------|-------|--------|--------------------|----------|-----------|----------|-------|
|                                    |                                           | Contact Hours   |           |       | Credit | Theory             |          | Practical |          | Total |
|                                    |                                           | Theory          | Practical | Total |        | Internal           | External | Term Work | External |       |
| CTUE501-<br>CTUE504<br>CTME503-504 | Elective-I                                | 3               | -         | 3     | 3      | 50                 | 50       | -         | -        | 100   |
| CTUC505                            | Advanced Java                             | 3               | 3         | 6     | 6      | 50                 | 50       | 50        | 50       | 200   |
| CTUC506                            | Web Development using .NET                | 3               | 3         | 6     | 6      | 50                 | 50       | 50        | 50       | 200   |
| CTUC507                            | Advanced Database Administration          | 3               | 3         | 6     | 6      | 50                 | 50       | 50        | 50       | 200   |
| CTUC508                            | Modern Software Engineering               | 2               | -         | 2     | 2      | 50                 | 50       | -         | -        | 100   |
| HSUS502                            | Foreign Languages                         |                 |           |       |        |                    |          | 25        | 25       | 50    |
| HSUS501                            | Academic Speaking and Presentation Skills | -               | 2         | 2     | 2      | -                  | -        | 25        | 25       | 50    |
|                                    |                                           | 14              | 11        | 25    | 25     | 500                |          | 450       |          | 950   |

**Elective-I**

| Course Code | Course Title                                 |
|-------------|----------------------------------------------|
| CTUE501     | Cloud Computing Essentials                   |
| CTUE502     | Mobile and Wireless Communications           |
| CTUE503     | Fuzzy Logic and Applications                 |
| CTUE504     | Distributed Operating Systems                |
| CTME503     | Introduction to Large Language Models (LLMs) |
| CTME504     | Cyber Security and Privacy                   |

**Teaching & Examination Scheme**  
**Master of Science in Information Technology (M.Sc. (IT)) Programme**  
 (Choice Based Credit System)

**Effective from Year 2025 - 26**  
**Semester - II**

| Course Code     | Course Title                            | Teaching Scheme |           |       |        | Examination Scheme |          |           |          |       |
|-----------------|-----------------------------------------|-----------------|-----------|-------|--------|--------------------|----------|-----------|----------|-------|
|                 |                                         | Contact Hours   |           |       | Credit | Theory             |          | Practical |          | Total |
|                 |                                         | Theory          | Practical | Total |        | Internal           | External | Term Work | External |       |
| CTUE509-CTUE512 | Elective-II                             | 3               | 3         | 6     | 6      | 50                 | 50       | 50        | 50       | 200   |
| CTUC513         | Web Development Using Open Source       | 3               | 3         | 6     | 6      | 50                 | 50       | 50        | 50       | 200   |
| CTUC514         | Advanced Mobile Application Development | 3               | 3         | 6     | 6      | 50                 | 50       | 50        | 50       | 200   |
| CTUC515         | Full Stack Web Development              | 3               | 3         | 6     | 6      | 50                 | 50       | 50        | 50       | 200   |
| HSUA501         | Academic Writing                        | -               | 2         | 2     | 2      | -                  | -        | 25        | 25       | 50    |
|                 | University Elective-I **                | -               | 2         | 2     | 2      | -                  | -        | 50        | 50       | 100   |
|                 |                                         | 12              | 16        | 28    | 28     | 400                |          | 550       |          | 950   |

\*\*Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **CAUD516-Internet and Web Designing** and **CAUD517-Mobile Application Development** course for others.

| Elective-II |                                             |
|-------------|---------------------------------------------|
| Course Code | Course Title                                |
| CTUE509     | Game Design                                 |
| CTUE510     | Advanced Python Programming                 |
| CTUE511     | Block Chain Programming                     |
| CTUE512     | HTTP Web Service for Enterprise Application |

**University Elective-I**

| <b>Sr. No.</b> | <b>Course Code</b> | <b>Course Name</b>                                                   | <b>Department/Faculty</b> |
|----------------|--------------------|----------------------------------------------------------------------|---------------------------|
| 1              | FTUD552            | Health and Safety Management                                         | Engineering               |
| 2              | FTUD502            | Sustainable Engineering Concepts and Life Cycle Analysis             | Engineering               |
| 3              | FTUD501            | Blockchain and its Applications                                      | Engineering               |
| 4              | OCMPH1003          | Introduction On Intellectual Property To Engineers And Technologists | Pharmacy                  |
| 5              | MBUD558            | Introduction to Operations Research                                  | Management                |
| 6              | CAUD518            | Software Project Management                                          | Computer Science          |
| 7              | FSUD551            | Energy resources, economics and sustainability                       | Applied Science           |
| 8              | FSUD552            | Nuclear Astrophysics                                                 | Applied Science           |
| 9              | FSUD553            | Computational Science in Engineering                                 | Applied Science           |
| 10             | FSUD554            | Plastic Waste Management                                             | Applied Science           |
| 11             | PTUD796            | Yoga and Positive Psychology for Managing Career and Life            | Physiotherapy             |
| 12             | NRMD551            | Mindfulness and Well-being: Living with Balance and Ease             | Nursing                   |

**Teaching & Examination Scheme**  
**Master of Science in Information Technology (M.Sc. (IT)) Programme**  
 (Choice Based Credit System)

**Effective from Year 2025 - 26**  
**Semester - III**

| Course Code     | Course Title                           | Teaching Scheme |           |       |        | Examination Scheme |          |           |          |       |
|-----------------|----------------------------------------|-----------------|-----------|-------|--------|--------------------|----------|-----------|----------|-------|
|                 |                                        | Contact Hours   |           |       | Credit | Theory             |          | Practical |          | Total |
|                 |                                        | Theory          | Practical | Total |        | Internal           | External | Term Work | External |       |
| CTUE601-CTUE603 | Elective-III                           | 3               | 3         | 6     | 6      | 50                 | 50       | 50        | 50       | 200   |
| CTUC604         | Software Quality Assurance and Testing | 3               | 3         | 6     | 6      | 50                 | 50       | 50        | 50       | 200   |
| CTUC605         | Mini Project                           | -               | 12        | 12    | 12     | -                  | -        | 200       | 200      | 400   |
| CLUV601         | Environmental Sciences                 | -               | 2         | 2     | 2      | -                  | -        | 25        | 25       | 50    |
|                 |                                        | 06              | 20        | 26    | 26     | 200                |          | 650       |          | 850   |

| Elective-III |                           |
|--------------|---------------------------|
| Course Code  | Course Title              |
| CTUE601      | Advanced Game Development |
| CTUE602      | Data Analytics            |
| CTUE603      | Internet of Things        |

**Teaching & Examination Scheme**  
**Master of Science in Information Technology (M.Sc. (IT)) Programme**  
(Choice Based Credit System)

**Effective from Year 2025 - 26**  
**Semester - IV**

| Course Code | Course Title                | Teaching Scheme |          |        | Internal              | End Semester Examination |                     | Total |       |
|-------------|-----------------------------|-----------------|----------|--------|-----------------------|--------------------------|---------------------|-------|-------|
|             |                             | Contact Hours   |          | Credit | Continuous Evaluation | Report                   | Presentation & Viva |       |       |
|             |                             | Inst.           | Industry |        |                       |                          |                     |       | Total |
| CTUC606     | Dissertation / Project Work | 2               | 28       | 30     | 30                    | 400                      | 200                 | 200   | 800   |



# TEACHING SCHEME & DETAILED SYLLABUS

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Faculty of Computer Science & Applications  
Smt. Chandaben Mohanbhai Patel Institute of  
Computer Applications

M.Sc. (IT) Programme  
(Choice Based Credit System)



**TEACHING SCHEME &  
DETAILED SYLLABUS**

**FOR**

**M.Sc. (IT) PROGRAMME  
(1<sup>st</sup> SEMESTER)**

**EFFECTIVE FROM  
ACADEMIC YEAR 2025-26**

**Teaching & Examination Scheme**  
**Master of Science in Information Technology (M. Sc. (IT)) Programme**  
**(Choice Based Credit System)**  
**Effective from Year 2025 - 26**  
**Semester – I**

| Course Code                        | Course Title                              | Teaching Scheme |           |       |        | Examination Scheme |          |           |          |       |
|------------------------------------|-------------------------------------------|-----------------|-----------|-------|--------|--------------------|----------|-----------|----------|-------|
|                                    |                                           | Contact Hours   |           |       | Credit | Theory             |          | Practical |          | Total |
|                                    |                                           | Theory          | Practical | Total |        | Internal           | External | Term Work | External |       |
| CTUE501-<br>CTUE504<br>CTME503-504 | Elective-I                                | 3               | -         | 3     | 3      | 50                 | 50       | -         | -        | 100   |
| CTUC505                            | Advanced Java                             | 3               | 3         | 6     | 6      | 50                 | 50       | 50        | 50       | 200   |
| CTUC506                            | Web Development using .NET                | 3               | 3         | 6     | 6      | 50                 | 50       | 50        | 50       | 200   |
| CTUC507                            | Advanced Database Administration          | 3               | 3         | 6     | 6      | 50                 | 50       | 50        | 50       | 200   |
| CTUC508                            | Modern Software Engineering               | 2               | -         | 2     | 2      | 50                 | 50       | -         | -        | 100   |
| HSUS502                            | Foreign Languages                         |                 |           |       |        | -                  |          | 25        | 25       | 50    |
| HSUS501                            | Academic Speaking and Presentation Skills | -               | 2         | 2     | 2      |                    | -        | 25        | 25       | 50    |
|                                    |                                           | 14              | 11        | 25    | 25     | 500                |          | 450       |          | 950   |

| Elective-I  |                                              |
|-------------|----------------------------------------------|
| Course Code | Course Title                                 |
| CTUE501     | Cloud Computing Essentials                   |
| CTUE502     | Mobile and Wireless Communications           |
| CTUE503     | Fuzzy Logic and Applications                 |
| CTUE504     | Distributed Operating Systems                |
| CTME503     | Introduction to Large Language Models (LLMs) |
| CTME504     | Cyber Security and Privacy                   |

## **CTUE501 - Cloud Computing Essentials**

### **(100 Marks)**

**Contact Hours: 03**

**Pre-requisite:** Operating System Concepts and Network Technology

**Methodology & Pedagogy:** During theory lectures the emphasis will be given on the fundamentals of cloud computing. Students will be introduced basic types, architecture, service providers, mechanism, security issues and some hidden aspects of cloud computing. Students will give practical exposure in form of case study and by showing cloud infrastructure of university.

**Outline of the course:**

| <b>Unit Number</b> | <b>Title of the Unit</b>                                 | <b>Minimum Number of Theory Hours</b> |
|--------------------|----------------------------------------------------------|---------------------------------------|
| <b>1</b>           | Evolution of Cloud Computing                             | <b>04</b>                             |
| <b>2</b>           | Understanding Cloud Computing and basic types            | <b>07</b>                             |
| <b>3</b>           | Fundamentals of Cloud Architecture and Service Providers | <b>07</b>                             |
| <b>4</b>           | Cloud Computing Mechanisms                               | <b>07</b>                             |
| <b>5</b>           | Cloud Computing Security and Business Use                | <b>06</b>                             |
| <b>6</b>           | Intensive analysis of Cloud Computing with case studies  | <b>05</b>                             |

**Total Hours (Theory): 36**

**Total Hours: 36**

**Detail Syllabus:**

**Unit I : Evolution of Cloud Computing** **Hours 04**

Introduction of Cloud Computing, Growth of technology, Paradigm Shift in Computing, distributed nature of service Provisioning, Support entrepreneurship using Cloud Computing.

**Unit II : Understanding Cloud Computing and basic types** **Hours 07**

Advantages and drawbacks of Cloud Computing, Essential component for Cloud contract, Major outage of Cloud Computing and Enhancers for Cloud Computing. Introduction to SaaS, PaaS, IaaS. Introduction to Public Cloud, Private Cloud, Hybrid Cloud and Community Cloud, Storage Services for Cloud Computing.

**Unit III : Fundamentals of Cloud Architecture and Service Providers** **Hours 07**

Workload Distribution Architecture, Resource Pooling Architecture, Dynamic Scalability Architecture, Elastic Resource Capacity Architecture, Service Load Balancing Architecture, Cloud Bursting Architecture, Elastic Disk Provisioning Architecture, Redundant Storage Architecture. Introduction to major Cloud Service Provider: Amazon Web Services, Google Cloud. Microsoft Windows Azure and Office 365, Hp Cloud, RackSpace, CSC Corp, Verizon Terramark, DropBox.

**Unit IV : Cloud Computing Mechanisms** **Hours 07**

Introduction to Cloud Infrastructure Mechanisms: Logical Network Perimeter, Virtual Server, Cloud Storage Device, Cloud Storage Levels, Network Storage Interfaces, Object Storage Interfaces, Database Storage Interfaces, Relational Data Storage, on-Relational Data Storage, Cloud Usage Monitor, Monitoring Agent, Resource Agent, Polling Agent, Resource Replication. Introduction to Cloud Management Mechanisms: Remote Administration System, Resource Management System, SLA Management System, Billing Management System.

**Unit V : Cloud Computing Security and Business Use** **Hours 06**

Introduction to Encryption, Symmetric Encryption, Asymmetric Encryption, Hashing, Digital Signature, Public Key Infrastructure (PKI), Identity and Access Management (IAM), Single SignOn (SSO), Cloud-Based Security Groups. Overview of Compliance and Certification, Access Control, Organizational Control. Benefits of Business using Cloud Computing, Risk of Cloud Computing, Cost factor in Cloud Computing.

**Unit VI : Intensive analysis of Cloud Computing with case studies      Hours 05**

Overview of Cloud services, Designing Solutions for the Cloud, Implement & Integrate Solutions, Emerging Markets and the Cloud

**Core Books:**

1. Kevin L. Jackson, Scott Goessling: Architecting Cloud Computing Solutions: Packt Publication: 2018
2. Thomas Erl, Zaigham Mahmood and Ricardo Puttini: Cloud Computing Concepts, Technology & Architecture, PHI, 2013.
3. S. Srinivasan: Cloud Computing Basics, Springer, 2014.

**Reference Books:**

1. Derrick Rountree, Ileana Castrillo : The Basics of Cloud Computing, Syngress, 2013.
2. Rajkumar Buyya, James Broberg, Andrzej M. Goscinsk: Cloud Computing- Principles and Paradigms, John Wiley & Sons, 2011.

**Web References:**

1. [http://whatisCloud.com/basic\\_concepts\\_and\\_terminology/Cloud](http://whatisCloud.com/basic_concepts_and_terminology/Cloud) [For basic terminology of Cloud Computing]
2. [http://www.tutorialspoint.com/Cloud\\_Computing/](http://www.tutorialspoint.com/Cloud_Computing/) [For cloud computing lecture notes]
3. <http://www.intel.in/content/dam/www/public/us/en/documents/guides/cloudcomputingvirtualization-building-private-iaas-guide.pdf> [For cloud computing virtualization]
4. [www.cs.purdue.edu/.../Any-Kim-Bhargava-MCCWorkshop.ppt](http://www.cs.purdue.edu/.../Any-Kim-Bhargava-MCCWorkshop.ppt) [Security issues PPTs]

**Course Outcomes:** Upon successful completion of the course, students will be,

|       |                                                                                     |
|-------|-------------------------------------------------------------------------------------|
| CO1 : | Students will learn basics of cloud computing                                       |
| CO2 : | The students will be familiar with various cloud architectures and services.        |
| CO3 : | They will get the knowledge of various network mechanics.                           |
| CO4 : | Students will get various business aspects of cloud computing with security aspects |
| CO5 : | They will be able to design and deploy Cloud Infrastructure                         |

**Course Outcomes Mapping:**

| Unit No. | Unit Name                                                | Course Outcomes |     |     |     |     |
|----------|----------------------------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                                          | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | Evolution of Cloud Computing                             | √               |     |     |     |     |
| 2        | Understanding Cloud Computing and basic types            |                 | √   |     |     |     |
| 3        | Fundamentals of Cloud Architecture and Service Providers |                 | √   |     |     |     |
| 4        | Cloud Computing Mechanisms                               |                 |     | √   |     |     |
| 5        | Cloud Computing Security and Business Use                |                 |     |     | √   |     |
| 6        | Hidden Aspects of Cloud Computing                        |                 |     |     |     | √   |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 3   | 1   | -   | -   | 1   | -   | 3   | 1   | 2   | 2    | -    | -    | 1    | -    |
| CO2 | 3   | 1   | 1   | 1   | 1   | 1   | 3   | 1   | 2   | 2    | 1    | 3    | 2    | -    |
| CO3 | 3   | 2   | 1   | 2   | 1   | 3   | 3   | 2   | 2   | 2    | -    | 3    | 2    | -    |
| CO4 | 3   | 3   | 2   | 3   | 1   | 3   | 3   | 2   | 2   | 2    | 3    | -    | 1    | -    |
| CO5 | 3   | 3   | 3   | 3   | 1   | 3   | 3   | 3   | 2   | 2    | 2    | -    | 3    | 1    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## **CTUE502 - Mobile and Wireless Communications**

### **(100 Marks)**

**Contact Hours: 03**

**Pre-requisite:** Basic of computer and its working

**Methodology & Pedagogy:** During the lecture sessions, the teacher will discuss the applications of wireless communication, underlying concepts of wireless communication, cellular network and its working. The students will also learn about 5g, edge computing and security issues in wireless communication.

#### **Outline of the course:**

| Unit Number | Title of the Unit                       | Minimum Number of Theory Hours |
|-------------|-----------------------------------------|--------------------------------|
| 1           | Introduction to wireless communications | 05                             |
| 2           | Antenna and wave propagation            | 07                             |
| 3           | Encoding and modulation techniques      | 05                             |
| 4           | Cellular wireless network               | 08                             |
| 5           | Wireless link improvement techniques    | 06                             |
| 6           | Advanced concepts in wireless networks  | 05                             |

**Total Hours (Theory): 36**

**Total Hours: 36**

## **Detail Syllabus:**

### **Unit I: Introduction to wireless communications**

**Hours 05**

Overview of communication system, component of communication system, digital vs analog communication, interface of analog to digital and digital to analog systems, conversation from analog to digital and digital to analog, examples of wireless communication, wireless vs. wired communication.

### **Unit II: Antenna and wave propagation**

**Hours 07**

Radiation patterns, antenna types, propagation modes, line of sight communication, attenuation and signal distortion, free space loss, noise, atmospheric absorption, multipath refraction.

### **Unit III: Encoding and modulation techniques**

**Hours 05**

Signal encoding criteria, digital data and analog signals – amplitude shift keying, frequency shift keying and phase shift keying, analog data and signal – amplitude modulation, angle modulation, analog data and digital signal – pulse code modulation, delta modulation.

### **Unit IV: Cellular wireless network**

**Hours 08**

Cellular system overview, organization, frequency reuse, increasing capacity, operation of cellular system, steps in a mobile switching center (MSC) controlled call between mobile users, mobile radio propagation effect, additional function in MSC controlled call, handoff performance metrics, handoff strategies used to determine instance of handoff, power control, traffic engineering.

### **Unit V: Wireless link improvement techniques**

**Hours 06**

Equalization, diversity error detection methods – parity check, cyclic redundancy check, block error correction code – block codes, hamming codes, cyclic codes, reed Solomon codes.

### **Unit VI: Advanced concepts in wireless concepts**

**Hours 05**

Introduction to Wi-Fi, WiMAX, ZigBee Networks, Wireless Adhoc Network, Overview of 4g and 5g cellular technologies, edge computing and fog computing, Security aspects of wireless computing.

## **Core Books:**

1. William Stallings: Wireless communication & networks: 3<sup>rd</sup> Edition: Pearson: 2012.
2. David Tse: Fundamentals of Wireless Communication: 2<sup>nd</sup> Edition: Cambridge Uni. Press: 2014
3. Aditya K. Jagannatham: Principles of modern wireless communication Systems Theory and Practice: 1<sup>st</sup> Edition: McGraw Hill:2017

**Reference Books:**

1. T.S.Rappaport, “Wireless Communications Principles and Practice”, 2nd edition, PHI,2002.
2. William C.Y.Lee, “Mobile Cellular Telecommunications Analog and Digital Systems”, 2<sup>nd</sup> edition, TMH, 1995

**Web References:**

1. [https://www.tutorialspoint.com/wireless\\_communication/index.htm](https://www.tutorialspoint.com/wireless_communication/index.htm) [Wireless networking tutorials]
2. <https://www.javatpoint.com/mobile-communication-tutorial> [Mobile computing tutorials]
3. <https://www.youtube.com/watch?v=f2wlHL1Sok8> [ Cellular system and wireless concepts -NPTEL course YouTube link]

**Course Outcomes:** Upon successful completion of the course, students will be,

|       |                                                                                                                               |
|-------|-------------------------------------------------------------------------------------------------------------------------------|
| CO1 : | Student will understand role of wireless networks in communication system.                                                    |
| CO2 : | Student will get insight into underlying principles of wireless communication system.                                         |
| CO3 : | The student will grasp concepts of error and flow control of wireless network.                                                |
| CO4 : | The student will get awareness of cellular network system and its working.                                                    |
| CO5 : | The student will know about modern and upcoming wireless communication technologies and concepts along with security aspects. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name                               | Course Outcomes |     |     |     |     |
|----------|-----------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                         | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | Introduction to wireless communications | √               |     |     |     |     |
| 2        | Antenna and wave propagation            |                 | √   |     |     |     |
| 3        | Encoding and modulation techniques      |                 |     | √   |     |     |
| 4        | Cellular wireless network               |                 |     |     | √   |     |
| 5        | Wireless link improvement techniques    |                 |     |     | √   |     |
| 6        | Advanced concepts in wireless concepts  |                 |     |     |     | √   |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 2   | 2   | 3   | 2   | 2   | 1   | 2   | 2   | 2   | -    | 1    | 1    | 3    | 2    |
| CO2 | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 2   | 3   | -    | 2    | 2    | 3    | 2    |
| CO3 | 2   | 2   | 3   | 3   | 2   | 2   | 3   | 2   | 2   | -    | 1    | 1    | 3    | 2    |
| CO4 | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 2   | 2   | 2    | 3    | 3    | 3    | 3    |
| CO5 | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 2   | 1    | 3    | 3    | 3    | 3    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## **CTUE503 - Fuzzy Logic and Applications**

### **(100 Marks)**

**Contact Hours: 03**

**Pre-requisite:** Basics of Set Theory, Fundamental of Computing

**Methodology & Pedagogy:** During theory lectures Basics of Fuzzy Logic, Fuzzy Set Theory, Fuzzy Relations, Fuzzification, Fuzzy systems and Applications will be introduced to students. Students will be made familiar with applications and systems related to Fuzzy Theory. Appropriate case studies will be discussed and assigned to students for real exposure related to Fuzzy logic.

**Outline of the course:**

| <b>Unit Number</b> | <b>Title of the Unit</b>                | <b>Minimum Number of Theory Hours</b> |
|--------------------|-----------------------------------------|---------------------------------------|
| 1                  | Introduction to Fuzzy Logic             | 05                                    |
| 2                  | Classical Sets and Fuzzy Sets           | 07                                    |
| 3                  | Classical Relations and Fuzzy Relations | 06                                    |
| 4                  | Fuzzification and Defuzzification       | 06                                    |
| 5                  | Fuzzy Systems and Classification        | 06                                    |
| 6                  | Fuzzy Logic - Applications              | 06                                    |

**Total Hours (Theory): 36**

**Total Hours: 36**

**Detail Syllabus:**

**Unit I : Introduction to Fuzzy Logic**

**Hours 05**

A Historical Perspective, The Utility of Fuzzy Systems, imitations of Fuzzy Systems, The Illusion: Ignoring, Uncertainty and Accuracy, Uncertainty and Information, The Unknown, Fuzzy Sets and Membership, Chance Versus Fuzziness, Sets as Points in Hypercubes.

**Unit II : Fuzzy Set Theory**

**Hours 07**

Classical Sets : Operations on Classical Sets, Properties of Classical (Crisp) Sets, Mapping of Classical Sets to Functions, Fuzzy Sets: Fuzzy Set Operations, Properties of Fuzzy Sets, Alternative Fuzzy Set Operations, Fuzzy Arithmetic.

**Unit III : Classical Relations and Fuzzy Relations**

**Hours 06**

Crisp Relations: Cardinality of Crisp Relations, Operations on Crisp Relations, Properties of Crisp Relations, Composition.

Fuzzy Relations: Cardinality of Fuzzy Relations, Operations on Fuzzy Relations, Properties of Fuzzy Relations, Fuzzy Cartesian Product and Composition, Fuzzy Intervals.

**Unit IV : Fuzzification and Defuzzification**

**Hours 06**

Features of the Membership Function, Various Forms, Fuzzification, Defuzzification to Crisp Sets and scalers, Approximate Reasoning, Fuzzy Inference, Fuzzy Logic in Databases, Information Retrieval with Fuzzy Logic, Quantification, Activity for choosing a suitable alternative needed for realizing goal.

**Unit V : Fuzzy Systems and Classification**

**Hours 06**

Fuzzy Logic - Control System, Adaptive Fuzzy Controller: Basic Steps for Implementing Adaptive Algorithm, Parameters for selecting an Adaptive Fuzzy Controller.

Fuzzy (Rule-Based) Systems, Graphical Techniques of Inference, Classification by Equivalence Relations, Crisp Relations, Fuzzy Relations, Fuzzy Classification: Hard c-Means (HCM), Fuzzy c-Means (FCM)

**Unit VI : Fuzzy Logic - Applications**

**Hours 06**

Fuzzy Logic - Applications in Pattern Recognition, Classification, Computer Network, Medical, Knowledge Engineering and Data Mining. Fuzzy logic with Neural Network. Intelligent control of complex systems. Exercises on Fuzzy Logic.

**Core Books:**

1. Timothy J. Ross: Fuzzy logic with Engineering Applications : Edition (4<sup>th</sup>), John Wiley & Sons, Ltd., 2017
2. Hans-Jurgen Zimmermann: Fuzzy Set Theory - and its Applications: Edition (4<sup>th</sup>), Springer Netherlands, 2001

**Reference Books:**

1. Klir, G., Yuan B. : Fuzzy Sets and Fuzzy Logic: Theory and Applications : Edition (4<sup>th</sup>), Pearson Education, 2015.
2. G. Chen and T. Pham : Introduction to fuzzy sets, fuzzy logic & Fuzzy control systems CRC Press, 2000.
3. Hung T. Nguyen, Elbert A. Walker: A First Course in Fuzzy Logic, CRC Press, 2005.
4. A. K. BHARGAVA: Fuzzy Set Theory Fuzzy Logic and their Applications S CHAND & Company Limited, 2013.
5. M. GANESH: INTRODUCTION TO FUZZY SETS AND FUZZY LOGIC, PHI Learning, 2006.
6. Barnabas Bede: Mathematics of Fuzzy Sets and Fuzzy Logic, Springer Berlin Heidelberg, 2012.

**Web References:**

1. <https://plato.stanford.edu/entries/logic-fuzzy/> [Fuzzy Logic]
2. <https://www.mathworks.com/help/fuzzy/what-is-fuzzy-logic.html> [What Is Fuzzy Logic?]
3. <https://www.guru99.com/what-is-fuzzy-logic.html> [Fuzzy Logic Tutorial: What is, Architecture, Application, Example]
4. [https://www.tutorialspoint.com/fuzzy\\_logic/fuzzy\\_logic\\_applications.htm](https://www.tutorialspoint.com/fuzzy_logic/fuzzy_logic_applications.htm) [Fuzzy Logic - Applications]

**Course Outcomes:** Upon successful completion of the course, students will be able

|       |                                                                                    |
|-------|------------------------------------------------------------------------------------|
| CO1 : | To understand concept of fuzzy logic principles.                                   |
| CO2 : | To understand the skill in basic understanding of fuzzy logic.                     |
| CO3 : | To explore the components of fuzzy set and fuzzy operations.                       |
| CO4 : | To explore the functional components of fuzzy logic classification and controller. |
| CO5 : | To explore applications related to fuzzy logic.                                    |

**Course Outcomes Mapping:**

| Unit No. | Unit Name                               | Course Outcomes |     |     |     |     |
|----------|-----------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                         | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | Introduction to Fuzzy Logic             | √               |     |     |     |     |
| 2        | Classical Sets and Fuzzy Sets           |                 | √   | √   |     |     |
| 3        | Classical Relations and Fuzzy Relations |                 |     | √   | √   |     |
| 4        | Fuzzification and Defuzzification       |                 |     | √   | √   |     |
| 5        | Fuzzy Systems and Classification        |                 |     |     | √   | √   |
| 6        | Fuzzy Logic - Applications              |                 |     |     |     | √   |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 3   | 2   | 2   | -   | -   | -   | 3   | -   | -   | -    | -    | 2    | 3    | 2    |
| CO2 | 3   | 3   | 2   | 2   | -   | -   | 3   | -   | -   | -    | -    | 2    | 3    | 2    |
| CO3 | 3   | 3   | 2   | 2   | 1   | -   | 3   | -   | -   | -    | -    | 2    | 3    | 2    |
| CO4 | 3   | 3   | 2   | 2   | 1   | -   | 3   | -   | -   | -    | -    | 2    | 3    | 2    |
| CO5 | 2   | 2   | 2   | 3   | 2   | -   | 3   | -   | -   | -    | -    | 2    | 3    | 2    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## **CTUE504 - Distributed Operating Systems**

### **(100 Marks)**

**Contact Hours: 03**

**Pre-requisite:** Operating Systems concepts

**Methodology & Pedagogy:** During theory lectures fundamental concept of Distributed Systems will be covered along with the issues and management of distributed memory like synchronization in Distributed Operating Systems, Resource Management, Handling file and database in distributed systems and the Security concerns.

#### **Outline of the Course:**

| <b>Unit No.</b> | <b>Title of the Unit</b>                                         | <b>Minimum Numbers of Hours Theory</b> |
|-----------------|------------------------------------------------------------------|----------------------------------------|
| 1               | Overview of Distributed Systems                                  | 07                                     |
| 2               | Distributed Shared Memory                                        | 07                                     |
| 3               | Synchronization in Distributed Operating Systems                 | 07                                     |
| 4               | Process and Resource Management in Distributed Operating Systems | 05                                     |
| 5               | Distributed File Systems                                         | 05                                     |
| 6               | Distributed Web based System and Security                        | 05                                     |

**Total Hours (Theory): 36**

**Total Hours: 36**

## **Detail Syllabus:**

### **Unit – I: Overview of Distributed Systems**

**Hours: 07**

Computer architecture: CICS, RISC, Multi-core, Computer networking: ISO/OSI Model, Evolution of operating systems, Introduction to Distributed operating system, issues in designing distributed operating system – transparency, Reliability, Flexibility, performance, scalability, heterogeneity, security, Emulation of existing operating system, Introduction to distributed computing environment, Introduction to distributed computing systems (DCS), DCS design goals, Transparencies, Fundamental issues.

### **Unit – II: Distributed Shared Memory**

**Hours: 07**

Introduction to RPC, RPC Model, Implementing RPC Mechanism, Stub Generation, RPC Message, Introduction to Distributed Shared Memory, General Architecture of DSM Systems, Design and implementation issues of DSM, Granularity, Structure of Shared memory space, Message passing communication, Transaction Communication, Group communication; Broadcast atomic protocols.

### **Unit – III: Synchronization in Distributed Operating Systems**

**Hours: 07**

Distributed Coordination, Temporal ordering of events, Lamport's logical clocks, Vector clocks; Ordering of messages, Physical clocks, Global state detection Clock Synchronization – Implementation, drifting of clocks, Clock synchronization issues, Clock Synchronization algorithms, Mutual Exclusion – Centralized approach, Distributed Approach, Token Passing Approach, Deadlocks in distributed systems, Load scheduling and balancing techniques.

### **Unit – IV: Process and Resource Management in Distributed Operating Systems**

**Hours: 05**

Introduction to resource management in distributed operating system, Desirable features of good global scheduling algorithm, Task Assignment approach, Load balancing Approach, Load Sharing Approach, Process migration – desirable feature of Process migration, process migration mechanisms, Process migration in heterogeneous systems, Advantages of process migration.

### **Unit – V: Distributed File and Database Systems**

**Hours: 05**

What is Distributed File system – Remote information sharing, User mobility, Availability, Diskless workstation. Types of services in distributed file system – Storage service, True file Service, Name Service. Desirable features of Distributed file system, File Models, File accessing models, Distributed database system.

**Unit – VI: Distributed Web based System and Security**

**Hours: 05**

Architecture, Processes, Communication, Naming, Synchronization, Consistency and Replication: Web Proxy Caching, Replication for Web Hosting Systems, Replication of Web Applications, Introduction of Security in Distributed OS, Overview of security techniques, Features, Need, Access Control, Security Management.

**Core Books:**

1. Tim Kindberg, George Coulouris, Jean Dollimore : Distributed Systems : Concepts & Design Paperback Pearson Education, 2010.
2. Pradip K. Sinha: Distributed Operating Systems Concepts and Design, Eastern Economy Edition, PHI, 2007.
3. Mukesh Singhal, Niranjana Shivaratri : Advanced Concepts In Operating Systems Paperback, McGraw Hill Education, 2017.

**Reference Books:**

1. Andrew S. Tanenbaum: Distributed Operating Systems, 5<sup>th</sup> Edition, Pearson, 2008.
2. Doreen L. Galli: Distributed Operating Systems Concepts and Practice, Prentice Hall, 2000.
3. Andrew S. Tanenbaum, Herbert Bos: Modern Operating Systems, 4<sup>th</sup> Edition, Pearson Education, 2014.

**Web References:**

1. [www.geeksforgeeks.org/types-of-operating-systems/](http://www.geeksforgeeks.org/types-of-operating-systems/) [Overview of Distributed OS]
2. [www.javatpoint.com/distributed-operating-system](http://www.javatpoint.com/distributed-operating-system) [Examples and application of Distributed OS]
3. [www.tutorialspoint.com/Distributed-Database-Management-System](http://www.tutorialspoint.com/Distributed-Database-Management-System) [ Database Management of Distributed OS]

**Course Outcomes:** Upon successful completion of the course, the students will:

|       |                                                                                              |
|-------|----------------------------------------------------------------------------------------------|
| CO1 : | Be familiar with Distributed Operating Systems and Shared Memory.                            |
| CO2 : | Be familiar with the Synchronization issues in Distributed Operating Systems                 |
| CO3 : | Be familiar with the Process and Resource management.                                        |
| CO4 : | Understand working of file and database in the distributed environment.                      |
| CO5 : | Understand the working of web based system and security concerns in the distributed systems. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name                                                        | Course Outcomes |     |     |     |     |
|----------|------------------------------------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                                                  | Co1             | Co2 | Co3 | Co4 | Co5 |
| 1        | Overview of Distributed Systems                                  | √               |     |     |     |     |
| 2        | Distributed Shared Memory and Remote Procedure Calls             | √               |     |     |     |     |
| 3        | Synchronization in Distributed Operating Systems                 |                 | √   |     |     |     |
| 4        | Process and Resource Management in Distributed Operating Systems |                 |     | √   |     |     |
| 5        | Distributed File and Database Systems                            |                 |     |     | √   |     |
| 6        | Distributed Web based System and Security                        |                 |     |     |     | √   |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 2   | 2   | 2   | 2   | 2   | 2   | 2   | 2   | 2   | 2    | 2    | 2    | 2    | 1    |
| CO2 | 2   | 3   | 2   | 3   | 3   | 2   | 2   | 2   | 2   | 2    | 2    | 2    | 2    | 1    |
| CO3 | 2   | 3   | 2   | 3   | 3   | 2   | 2   | 2   | 2   | 2    | 2    | 2    | 2    | 2    |
| CO4 | 3   | 2   | 2   | 2   | 3   | 2   | 2   | 2   | 2   | 2    | 2    | 2    | 2    | 2    |
| CO5 | 3   | 2   | 2   | 2   | 3   | 2   | 2   | 2   | 2   | 2    | 2    | 2    | 2    | 2    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## CTME503 - Introduction to Large Language Models (LLMs)

### Description:

### Credit and Week:

| Teaching Scheme | Week | Marks | Credit |
|-----------------|------|-------|--------|
|                 | 12   | 100   | 3      |

### About the course:

This course introduces the fundamental concepts underlying Large Language Models (LLMs). It starts with an introduction to the various problems in NLP, and discusses how to approach the problem of language modeling using deep learning. It describes the architectural intricacies of Transformers and the pre-training objectives of the different Transformer-based models. It also discusses the recent advances in LLM research, including LLM alignment, prompting, parameter-efficient adaptation, hallucination, bias and ethical considerations. This course prepares a student to comprehend, critique and approach various research problems on LLMs.

### Pre-requisites:

Machine Learning, Python Programming Optional: Deep Learning, Machine Learning, Python Programming Optional: Deep Learning

Introduction to Machine Learning: <https://nptel.ac.in/courses/106105152>

Python for Data Science: <https://nptel.ac.in/courses/106106212>

### Industry support:

All those industries whose work involves machine learning, such as Google, Microsoft, Adobe, IBM, Accenture, JP Morgan, Wipro, Flipkart, Amazon, etc.

### Course layout:

#### Week 1

1. Course Introduction.
2. Introduction to NLP (NLP Pipeline, Applications of NLP).

#### Week 2

1. Introduction to Statistical Language Models.
2. Statistical Language Models: Advanced Smoothing and Evaluation.

### **Week 3**

1. Introduction to Deep Learning (Perceptron, ANN, Backpropagation, CNN).
2. Introduction to PyTorch.

### **Week 4**

1. Word Representation.
  - a. Word2Vec, fastText
  - b. GloVe
2. Tokenization Strategies.

### **Week 5**

1. Neural Language Models.
  - a. CNN, RNN
  - b. LSTM, GRU
2. Sequence-to-Sequence Models, Greedy Decoding, Beam search.
3. Other Decoding Strategies: Nucleus Sampling, Temperature Sampling, Top-k Sampling.
4. Attention in Sequence-to-Sequence Models.

### **Week 6**

1. Introduction to Transformers.
  - a. Self and Multi-Head Attention
  - b. Positional Encoding and Layer Normalization.
2. Implementation of Transformers using PyTorch.

### **Week 7**

1. Pre-Training Strategies: ELMo, BERT (Encoder-only Model).
2. Pre-Training Strategies: Encoder-decoder and Decoder-only Models.
3. Introduction to HuggingFace.

### **Week 8**

1. Instruction Tuning.
2. Prompt-based Learning.
3. Advanced Prompting Techniques and Prompt Sensitivity.
4. Alignment of Language Models with Human Feedback (RLHF).

### **Week 9**

1. Open-book question answering: The case for retrieving from structured and unstructured sources; retrieval-augmented inference and generation.
2. Retrieval augmentation techniques.
  - a. Key-value memory networks in QA for simple paths in KGs
  - b. Early HotPotQA solvers, pointer networks, reading comprehension
  - c. REALM, RAG, FiD, Unlimiformer

d. KGQA (e.g., EmbedKGQA, GrailQA)

### **Week 10**

1. Knowledge graphs (KGs)
  - a. Representation, completion
  - b. Tasks: Alignment and isomorphism
  - c. Distinction between graph neural networks and neural KG inference.

### **Week 11**

1. Parameter-efficient Adaptation (Prompt Tuning, Prefix Tuning, LoRA)
2. An Alternate Formulation of Transformers: Residual Stream Perspective
3. Interpretability Techniques.

### **Week 12**

1. Overview of recently popular models such as GPT-4, Llama-3, Claude-3, Mistral, and Gemini
2. Ethical NLP – Bias and Toxicity.
3. Conclusion.

### **Books and references**

1. Tanmoy Chakraborty, Introduction to Large Language Models, Wiley India, 1st Edition, 2025. ISBN : 9789363864740
2. Dan Jurafsky and James H. Martin, Speech and Language Processing, 2nd edition, Pearson Press, 2008.
3. Jacob Eisenstein, Natural Language Processing, First edition, The MIT Press, 2019.

### **Criteria to get a certificate:**

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

***YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE  $\geq 10/25$  AND EXAM SCORE  $\geq 30/75$ . If one of the 2 criteria is not met, you will not get the certificate even if the Final score  $\geq 40/100$ .***

## CTME504 - Cyber Security and Privacy

### Description:

### Credit and Week:

| Teaching Scheme | Week | Marks | Credit |
|-----------------|------|-------|--------|
|                 | 12   | 100   | 3      |

### About the course:

This course introduces to management students the concepts, technologies, practices and challenges associated with cybersecurity as applied in organizations. Protection as well as disclosure of information pose unique challenges and also allude to economic and technological implications. The course takes a broad view of cyber security along with information privacy by analysing relevant organizational, human, legal and policy issues. Through this course, students will explore cyber security along with information privacy with a managerial focus. Students are expected to develop a wholesome understanding about cyber security and privacy risks to businesses covering governance, compliance and risk mitigation and closely study certain business domains.

### Pre-requisites:

A core course on Management Information Systems desirable (not mandatory).

### Industry support:

IT services industry, AI/Block chain start ups, Industry 4.0, autonomous vehicles industry.

Week 1 : Foundations, cyber security, information security and related concepts, Principles of information security management, Confidentiality, Integrity, Availability and related concepts.

Week 2 : Security management, Governance, Risk and Compliance (GRC) , Contingency planning, incidence response, disaster recovery and business continuity.

Week 3 : Understanding security policy, security behavior , Risk management: Risk identification, threat modeling, strategies.

Week 4 : Control strategies and protection mechanisms (Guest lecture), Cryptography for security.

Week 5 : Information security and privacy, Regulatory landscape: Fair information practices, US regulatory frameworks .

Week 6 : Regulatory landscape: EU's GDPR and its implications and other privacy and cyber security regulations, Cyber security and privacy in the Indian context, evolution and issues.

Week 7 : Economics of privacy, privacy calculus and trade-offs, privacy paradox, Managing stakeholders, making choices on security and privacy Course layout:

### Books and references

- Michael E. Whitman, Herbert J. Mattord, (2018). Principles of Information Security, 6th edition, Cenage Learning, N. Delhi.
- Darktrace, "Technology" <https://www.darktrace.com/en/technology/#machine-learning>, accessed November 2018.
- Van Kessel, P. Is cyber security about more than protection? EY Global Information Security Survey 2018-2019.
- Johnston, A.C. and Warkentin, M. Fear appeals and information security behaviors: An empirical study. MIS Quarterly, 2010.
- Arce I. et al. Avoiding the top 10 software security design flaws. IEEE Computer Society Center for Secure Design (CSD), 2014.
- Smith, H. J., Dinev, T., & Xu, H. Information privacy research: an interdisciplinary review. MIS Quarterly, 2011.
- Subramanian R. Security, privacy and politics in India: a historical review. Journal of Information Systems Security (JISSec), 2010.
- Acquisti, A., John, L. K., & Loewenstein, G. **What** is privacy worth? The Journal of Legal Studies, 2013
- Xu H., Luo X.R., Carroll J.M., Rosson M.B. The personalization privacy paradox: An exploratory study of decision making process for location-aware marketing. Decision Support Systems, 2011.

### Criteria to get a certificate:

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

***YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE  $\geq 10/25$  AND EXAM SCORE  $\geq 30/75$ . If one of the 2 criteria is not met, you will not get the certificate even if the Final score  $\geq 40/100$ .***

## CTUC505 - Advanced Java (200 Marks)

**Contact Hours: 06**

**Pre-requisite:** Fundamental Knowledge of Core Java (Java SE) concepts.

**Methodology & Pedagogy:** Advanced Java fundamentals like JDBC, Servlet, JSP and Servlet Security provides capabilities and features for developing web application which is essential for developer in order to understand entire web application development life cycle. During the theory sessions of the syllabus, fundamentals of the mentioned technologies will be explained in detail with real time examples. Moreover, practical session will provide hands-on exposure about JDBC, Servlet, JSP and Servlet Security with variety of practical assignment. Also, MVC architecture using these technologies will be covered during theory as well as practical sessions.

### Outline of the course:

| Unit Number | Title of the Unit                        | Minimum Number of Hours |           |
|-------------|------------------------------------------|-------------------------|-----------|
|             |                                          | Theory                  | Practical |
| 1           | JDBC Programming                         | 06                      | 36        |
| 2           | Web Application Components Fundamentals  | 05                      |           |
| 3           | Servlet API                              | 08                      |           |
| 4           | Advanced Servlet Features and Security   | 05                      |           |
| 5           | Java Server Pages                        | 07                      |           |
| 6           | Packaging and Deploying Web Applications | 05                      |           |

**Total Hours (Theory): 36**

**Total Hours (Lab): 36**

**Total Hours: 72**

## **Detail Syllabus:**

### **Unit I: JDBC Programming**

**Hours 06**

The JDBC Connectivity Model, Database Programming: Connecting to the Database, creating a SQL Query, Getting the Results, Updating Database Data, Error Checking and the SQLException Class, the SQLWarning class, the Statement Interface, PreparedStatement, CallableStatement, the ResultSet Interface, Updatable Result sets, JDBC Types, Executing SQL Queries, ResultSetMetaData, Executing SQL Updates, Transaction Management

### **Unit II: Web Application Components Fundamentals**

**Hours 05**

Basics of web application development and web components, Introduction of MVC design pattern, Fundamentals of: Containers, Packaging Web Applications, Web Application Structure, JAR Files, WAR Files, HTTP, HTTP request methods, HTML Form Processing, HTTP request response cycle.

### **Unit III: Servlet API**

**Hours 08**

Servlet Model: Introducing Servlet, Introducing Servlet & the MVC Pattern, Introducing javax.servlet Packages, Introducing HTTP & Servlets, Understanding the Request/Response Cycle, Input & Output Streams, Introducing Servlet/ Container Communication, Introducing ServletContext and ServletConfig, RequestDispatcher Interface. The Filter API: Filter, Filter Chain and Filter Config

### **Unit IV: Advanced Servlet Features and Security**

**Hours 05**

Cookies and Session Management: Understanding state and session, Understanding Session Timeout and Session Tracking, URL Rewriting. Understanding the Stateless nature of HTTP, Why Track Client Identity & State Maintain Sessions, Session Management Using the Servlet API.

### **Unit V: Java Server Pages**

**Hours 07**

JSP Overview: The Problem with Servlets, Life Cycle of JSP Page, JSP Processing, JSP Application Design with MVC, Setting Up the JSP Environment JSP Directives, JSP Action, JSP Implicit Objects JSP Form Processing, JSP Session and Cookies Handling, JSP Session Tracking JSP Database Access, JSP Standard Tag Libraries, JSP Custom Tag, JSP Expression Language, JSP Exception Handling, JSP XML Processing

### **Unit VI: Packaging and Deploying Web Applications**

**Hours 05**

Basics of server container, build a WAR artifact for the web application, configure the Docker connection node and deploy the application to server container

**Core Books:**

1. Cay S Horstmann, Gary Cornell: Core Java, Volume II – Advanced Features, 8th Edition, Pearson Education.
2. Marty Hall, Larry Brown: Core Servlets and JavaServer Pages, Volume 2, Advanced Technologies, 2nd Edition, Pearson Education, 2008.

**Reference Books:**

1. Bryan Basham, Kathy Sierra and Bert Bates: Head First Servlet and JSP, O'Reilly Publication, 1st Edition
2. James Keogh: Complete Reference J2EE, Mcgraw publication

**Web References:**

1. <http://courses.coreservlets.com/Course-Materials/csajsp2.html> [Servlet Basics]
2. <http://www.ceit.es/assignaturas/InteInfo/Recursos/Servlets/JavaServlets.pdf> [Servlet Tutorial PDF]
3. <http://www.msuniv.ac.in/AdvancedJavaProgrammingwithDatabaseApplication.pdf> [JDBC Tutorial]
4. [www.doc.ic.ac.uk/~rcheung/teaching/2720/ppt/lecture12.ppt](http://www.doc.ic.ac.uk/~rcheung/teaching/2720/ppt/lecture12.ppt) [JSP Tutorial Slides]

**Course Outcomes:** Upon successful completion of the course, students will:

|       |                                                                                      |
|-------|--------------------------------------------------------------------------------------|
| CO1 : | Able to connect Java application with various RDBMS using JDBC                       |
| CO2 : | Have understanding of web application architecture and terminologies                 |
| CO3 : | Able to develop real time web application using Servlet and JSP technologies         |
| CO4 : | Be familiar with the concepts of JSPs, Servlets and related security issues.         |
| CO5 : | Apply Model-View-Controller architecture to build complex client-server applications |

**Course Outcomes Mapping:**

| Unit No. | Unit Name                                | Course Outcomes |     |     |     |     |
|----------|------------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                          | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | JDBC Programming                         | √               |     | √   |     |     |
| 2        | Web Application Components Fundamentals  |                 | √   | √   |     |     |
| 3        | Servlet API                              |                 |     | √   | √   | √   |
| 4        | Advanced Servlet Features and Security   |                 |     | √   | √   | √   |
| 5        | Java Server Pages                        |                 |     | √   | √   | √   |
| 6        | Packaging and Deploying Web Applications |                 |     |     |     | √   |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 2   | 1   | 2   | 1   | 1   | -   | -   | -   | -   | -    | -    | -    | 3    | 2    |
| CO2 | 3   | 2   | 3   | 2   | -   | -   | -   | -   | -   | -    | 2    | 1    | 2    | 2    |
| CO3 | 3   | 2   | 3   | 2   | 2   | -   | -   | -   | -   | 2    | 2    | 3    | 3    | 3    |
| CO4 | 3   | 2   | 3   | 2   | 2   | 2   | -   | -   | 1   | -    | 2    | 1    | 3    | 2    |
| CO5 | 2   | 2   | 3   | 3   | 2   | -   | 1   | 1   | -   | 2    | 3    | 3    | 3    | 3    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## CTUC506 - Web Development Using .NET

### (200 Marks)

**Contact Hours: 06**

**Pre-requisite:** Familiarity with basic concepts of object oriented programming

**Methodology & Pedagogy:** In the lectures students will learn about basics of .NET framework, ASP.NET and other advanced concepts related to programming with .NET framework. In the lab sessions the student will learn about solving real world problems with .NET framework and related technologies.

**Outline of the course:**

| Unit Number | Title of the Unit                     | Minimum Number of Hours |           |
|-------------|---------------------------------------|-------------------------|-----------|
|             |                                       | Theory                  | Practical |
| 1           | Introduction to ASP.NET               | 05                      | 36        |
| 2           | Applications development with ASP.NET | 06                      |           |
| 3           | Database with ADO.Net                 | 08                      |           |
| 4           | MVC architecture in .NET framework    | 08                      |           |
| 5           | SOA in .NET framework                 | 05                      |           |
| 6           | AJAX in .NET                          | 04                      |           |

**Total Hours (Theory): 36**

**Total Hours (Lab): 36**

**Total Hours: 72**

## **Detail Syllabus:**

### **Unit I: Introduction to ASP.NET**

**Hours 05**

Introduction to .NET framework, Introduction to Web Programming, Introduction to ASP and ASP.NET, Deploying ASP.NET Application

### **Unit II: Applications development with ASP.NET**

**Hours 06**

ASP.NET Page Life Cycle, Structure of an ASP.NET Page: ASPX Page, Code behind File, WebConfig and machine config, Developing a web form, Working with master pages, State management techniques, Application Tracing, Error Handling

### **Unit III: Database with ADO.Net**

**Hours 08**

Introduction to ADO and ADO.NET, ADO.NET architecture, Connection Oriented and Connection Less architecture, Binding data to web controls and working with data controls, working with XML

### **Unit IV: MVC architecture in .NET framework**

**Hours 08**

Introduction to MVC Architecture in ASP.NET, CRUD operation using MVC architecture, working with web controls in MVC architecture, introduction to LINQ.

### **Unit V: SOA in .NET framework**

**Hours 05**

Introduction to service oriented architecture concept, SOAP protocol, Concept of UDDI and WSDL, Applications of SOA, Building SOA application with ASP.NET.

### **Unit VI: AJAX in .NET**

**Hours 04**

Introduction to ASP.NET AJAX, AJAX control toolkit and extender controls.

## **Core Books:**

1. Cogent Solutions Inc.: ASP.Net 4.5 Black book, Dreamtech press, 2019
2. Mridila Parihar, Essam Ahmed: ASP .Net Bible, Wiley, 2017
3. Stephon Walther: ASP.Net Unleashed, BPB publication, 2018

## **Reference Books:**

1. Mesbah Ahmed, Chris Garrett, Jeremy Faircloth, Chris Payne: ASP.Net Programming. Developer's Guide, Dreamtech, 2016.
2. Greg Buczek: ASP.Net Tips & Techniques, Tata McGraw Hill Edition, 2016.

3. Bolton, Justin Langford, Glenn Berry, Gavin Payne, Amit Banerjee, Rob Farley: Professional SQL Server 2019 internals and troubleshooting, Wiley India publication, October, 2019.

**Web References:**

1. <http://msdn.microsoft.com/en-us/aa336522.aspx> [For software download]
2. <https://docs.microsoft.com/en-us/aspnet/overview> [For ASP.NET technical documentation]
3. [https://www.tutorialspoint.com/dotnet\\_core/dotnet\\_core\\_overview.htm](https://www.tutorialspoint.com/dotnet_core/dotnet_core_overview.htm) [for .NET Core]

**Course Outcomes:** Upon successful completion of the course, students will be,

|       |                                                                                          |
|-------|------------------------------------------------------------------------------------------|
| CO1 : | Students will be able to understand .NET framework and develop ASP.NET application.      |
| CO2 : | Students will be able to develop database centric applications.                          |
| CO3 : | Students will be familiar with concepts of MVC for application development.              |
| CO4 : | Students will able to understand and develop SOA based applications with .NET framework. |
| CO5 : | Students will be familiar with the concepts of AJAX                                      |

**Course Outcomes Mapping:**

| Unit No. | Unit Name                             | Course Outcomes |     |     |     |     |
|----------|---------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                       | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | Introduction to ASP.NET               | √               |     |     |     |     |
| 2        | Applications development with ASP.NET | √               |     |     |     |     |
| 3        | Database with ADO.Net                 |                 | √   |     |     |     |
| 4        | MVC architecture in .NET framework    |                 |     | √   |     |     |
| 5        | SOA in .NET framework                 |                 |     |     | √   |     |
| 6        | AJAX in .NET                          |                 |     |     |     | √   |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 3   | 3   | 3   | 3   | -   | 2   | 1   | -   | -   | 2    | -    | -    | 2    | 1    |
| CO2 | 2   | 3   | 3   | 2   | -   | 2   | 1   | -   | -   | 2    | 2    | 1    | 3    | 1    |
| CO3 | 1   | 2   | 2   | -   | -   | -   | -   | -   | -   | -    | 2    | 1    | 3    | 2    |
| CO4 | 2   | 3   | 3   | -   | 2   | -   | -   | -   | -   | -    | 2    | 1    | 3    | 2    |
| CO5 | 3   | 2   | 3   | 2   | -   | -   | 3   | -   | -   | -    | -    | 2    | 3    | 3    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## CTUC507 - Advanced Database Administration (200 Marks)

**Contact Hours: 06**

**Pre-requisite:** To be able to implement a simple DBMS and design and implement information systems using DBMS technology.

**Methodology & Pedagogy:** During theory lectures, illustrations of certain complex real world applications, which emphasis the use of advanced concepts of databases, will be discussed. The fundamental concepts regarding database development activities, various database management systems and other advanced issues in database management will also be discussed. In addition, there may be announced or unannounced quizzes/assignments. During Practical sessions, students will be required to carry out case studies using the concepts and techniques they have learnt during theory sessions.

**Outline of the course:**

| Unit No. | Title of the Unit                     | Minimum Number of Hours |           |
|----------|---------------------------------------|-------------------------|-----------|
|          |                                       | Theory                  | Practical |
| 1        | Introduction to DBA                   | 05                      | 36        |
| 2        | User Access and Security              | 06                      |           |
| 3        | Data Storage Management and Migration | 07                      |           |
| 4        | Managing Database Backup and Recovery | 07                      |           |
| 5        | Database Security and Auditing        | 05                      |           |
| 6        | Query Processing and Optimization     | 06                      |           |

**Total Hours (Theory): 36**

**Total Hours (Lab): 36**

**Total Hours: 72**

## **Detail Syllabus:**

### **Unit I: Introduction to DBA**

**Hours: 05**

Types of Oracle Database Users, DBA Roles and Responsibilities; Database Architecture; ORACLE logical and physical database structure; Memory and Process Structure, Installation and configuration of a Database Server, SQLPLUS Overview, creating a database.

### **Unit II: User Access and Security**

**Hours: 06**

Administrative User Accounts, Database Administrator Authentication, Administrative Privileges and its operations, managing Password authentication, Granting and Revoking Administrative Privileges.

### **Unit III: Data Storage Management and Migration**

**Hours: 07**

Oracle Multitenant Architecture – Concept of CDBs and PDBs, Disk Storage, Basic File structure and Indexing, Indexing structure for Files, Data Migration, Data Utilities (SQL loader and Import – Export)

### **Unit IV: Managing Database Backup and Recovery**

**Hours: 07**

Backup and Recovery Overview, Database backup, restoration and recovery, defining a backup and recovery strategy, Backup and Recovery options; Data Dump; User-Managed Backup and Recovery; Configuring RMAN; RMAN Backups, Restore and Recovery.

### **Unit V: Database Security and Auditing**

**Hours: 05**

Database Security and Auditing; Database Authentication and Authorization Methods; Data Encryption Techniques, Virtual Private Database

### **Unit VI: Query Processing and Optimization**

**Hours: 06**

Query Processing, Steps of Query Processing, Query Optimization, Distributed Database System, Centralized Database Management System, Difference of Distributed Database System and Centralized Database System, Advantage and Disadvantage of Distributed and Centralized Database Management System.

## **Core Books:**

1. Oracle® Database Database Administrator's Guide by Mark Doran, Padmaja Potineni, Rajesh Bhatiya, Oracle Press.
2. Fundamentals of Database System by Ramez Elmasri and Shamkant B Navathe, Pearson Edition.

3. Oracle Database 19c DBA by Examples: Installation and Administration by Ravinder Gupta, Kindle Edition.

**Reference Books:**

1. SQL, PL/SQL - The Programming language of Oracle by Ivan Bayross, BPB Publications.
2. Database concepts - SQL and PL/SQL Programming by Dr. Prashant M Dolia, Mr. Kalpesh U Gundigara and Ms. Vibha H Mehta, Nirav Prakashan.
3. SQL and PL/SQL Practice Book with solutions by Anjali Jivani and Amisha Shingala, Nirav and Rupal Prakashan.

**Web References:**

1. A Practical Guide to Oracle Database Administration for DBA & Developers (oracletutorial.com)

**Course Outcomes:** Upon successful completion of the course, students will be,

|       |                                                                                                    |
|-------|----------------------------------------------------------------------------------------------------|
| CO1 : | Installation and configuration of a Database Server including creation and management of databases |
| CO2 : | Be familiar with database storage management                                                       |
| CO3 : | Manage backup and recovery                                                                         |
| CO4 : | Control user security                                                                              |
| CO5 : | Managing database query optimization                                                               |

**Course Outcomes Mapping: -**

| Unit No. | Unit Name                             | Course Outcomes |     |     |     |     |
|----------|---------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                       | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | Introduction to DBA                   | √               |     |     |     |     |
| 2        | User Access and Security              | √               | √   |     |     |     |
| 3        | Data Storage Management and Migration | √               | √   | √   |     |     |
| 4        | Managing Database Backup and Recovery |                 | √   |     | √   | √   |
| 5        | Database Security and Auditing        |                 |     | √   |     | √   |
| 6        | Query Processing and Optimization     |                 |     |     | √   | √   |

**Course Articulation Matrix: -**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|
| CO1 | 3   | 3   | 1   | 3   | 1   | 3   | 1   | 2   | -   | -    | -    | 3    | -    |
| CO2 | 2   | 2   | 3   | 2   | 1   | 3   | 3   | 2   | -   | -    | -    | 3    | -    |
| CO3 | 3   | 3   | 2   | 1   | 2   | 3   | 2   | 1   | -   | -    | -    | 3    | 3    |
| CO4 | 3   | 3   | 3   | 1   | 2   | 3   | 3   | 1   | -   | -    | -    | 3    | 3    |
| CO5 | 3   | 3   | 2   | 3   | 2   | 2   | 3   | 1   | -   | -    | -    | 3    | 3    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## CTUC508 - Modern Software Engineering (100 Marks)

**Contact Hours: 02**

**Pre-requisite:** Analysis and design of Information Systems, Fundamental concepts of ObjectOriented Programming.

**Methodology & Pedagogy:** During theory sessions, the focus will be given on different concepts of softwareengineering and related aspects such as Project management and Agile software development. Theemphasis will be given on the basics of software engineering concepts and UML. Students will get exposure to various concepts of DevOps, as well as Process Improvement and Reengineering.

### Outline of the course:

| Unit Number | Title of the Unit                           | Minimum Number of TheoryHours |
|-------------|---------------------------------------------|-------------------------------|
| 1           | Introduction to Software Engineering        | 04                            |
| 2           | Requirement analysis and project management | 04                            |
| 3           | Understanding UML                           | 04                            |
| 4           | Agile Development                           | 05                            |
| 5           | Overview of DevOps                          | 04                            |
| 6           | Software maintenance and risk management    | 03                            |

**Total Hours (Theory): 24**

**Total Hours:24**

**Detail Syllabus:**

**Unit I: Introduction to Software Engineering** **Hours 04**

The Role of Software in today's era, Software Engineering: Software Process Models, The Linear Sequential Model, The Prototyping Model, Evolutionary Process Models

**Unit II: Requirement analysis and project management** **Hours 04**

Requirement Specification (SRS), Requirement Analysis and Requirement Elicitation, Requirement Engineering. Introduction to the basic concepts of Project Scheduling, Project scheduling principles, relationships, Process Definition, Effort Estimation and Scheduling, Project Management Plan, Configuration Management.

**Unit III: Understanding UML** **Hour 04**

Introduction to UML, Structure and Unstructured modelling: Class Diagram, Use case Diagram, Interaction Diagram, Activity Diagram.

**Unit IV: Agile Development** **Hours 05**

Introduction to Agile Software Development, Characteristics of Agile Process, Problems with Agile methods, Extreme Programming, Scrum.

**Unit V: Overview of DevOps** **Hour 05**

What is DevOps?, DevOps Importance and Benefits, DevOps Principles and Practices, 7 C's of DevOps Lifecycle for Business Agility, Challenges with DevOps Implementation.

**Unit VI: Software maintenance and risk management** **Hour 03**

Software risks, risk identification, risk projection, risk mitigation, monitoring and management. Concept of software reengineering, reverse engineering, Applications of Software Engineering.

**Core Books:**

1. Roger S.Pressman, Software Engineering- A practitioner's Approach, McGraw-Hill International Editions
2. Grady Booch, James Rumbaugh, Ivar Jacobson: The Unified Modeling Language User
3. Rajib Mall, Fundamentals of software Engineering, Prentice Hall of India.

**Reference Books:**

1. Ian Sommerville, Software engineering, Pearson education Asia
2. Pankaj Jalote, Software Engineering – A Precise Approach Wiley
3. Merlin Dorfman (Editor), Richard H. Thayer (Editor) ,Software Engineering
4. Robert C. "Uncle Bob" Martin , Clean Architecture: A Craftsman's Guide to Software Structure and Design
5. Deepak Gaikwad, Viral Thakkar, DevOps Tools from Practitioner's ViewPoint, Wiley

**Web References:**

1. [https://en.wikibooks.org/wiki/Introduction\\_to\\_Software\\_Engineering/Process/Methodology](https://en.wikibooks.org/wiki/Introduction_to_Software_Engineering/Process/Methodology) [Introduction to Software Engineering/Process/Methodology]
2. <https://www.uml-diagrams.org/> [UMLUnits]
1. <https://nptel.ac.in/courses/106/101/106101061/> [Agile Software Development and Extreme Programming and Overview of DevOps]

**Course Outcomes:** Upon successful completion of the course, students will be,

|       |                                                              |
|-------|--------------------------------------------------------------|
| CO1 : | Understanding Software Engineering Process Models            |
| CO2 : | Able to analyze requirements and manage project              |
| CO3 : | Get familiar with the concepts of Unified Modeling Language  |
| CO4 : | Understanding of the Agile software development concepts     |
| CO5 : | Able to apply DevOps concepts and understand risk management |

### Course Outcomes Mapping:

| Unit No. | Unit Name                                   | Course Outcomes |     |     |     |     |
|----------|---------------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                             | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | Introduction to Software Engineering        | √               |     |     |     |     |
| 2        | Requirement analysis and project management |                 | √   |     |     |     |
| 3        | Understanding UML                           |                 |     | √   |     |     |
| 4        | Agile Development                           |                 |     |     | √   |     |
| 5        | Overview of DevOps                          |                 |     |     |     | √   |
| 6        | Software maintenance and risk management    |                 |     |     |     | √   |

### Course Articulation Matrix:

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 2   | 2   | 3   | 2   | 2   | 3   | 3   | 2   | 2   | 2    | 1    | 2    | 3    | 3    |
| CO2 | 3   | 2   | 2   | 2   | 2   | 3   | 3   | 3   | 3   | 2    | 3    | 2    | 3    | 3    |
| CO3 | 2   | 2   | 3   | 2   | 2   | 2   | 2   | 2   | 2   | 1    | 1    | 2    | 3    | 2    |
| CO4 | 3   | 3   | 3   | 2   | 2   | 2   | 2   | 1   | 1   | 1    | 3    | 2    | 3    | 2    |
| CO5 | 2   | 2   | 3   | 2   | 1   | 3   | 2   | 1   | 1   | 1    | 2    | 1    | 3    | 2    |

## HSUS502 - FOREIGN LANGUAGES (French)

### Credits and Hours:

| Teaching Scheme | Theory | Practical | Tutorial | Total | Credit    |
|-----------------|--------|-----------|----------|-------|-----------|
| Hours/week      | --     | 02/01     | --       | 30/15 | <b>02</b> |
| Marks           | --     | 100       | --       | 100   |           |

### Pre-requisite courses:

- French Language Studies- Introduction (Coursera)

### Outline of the Course:

| Sr. No. | Title of the unit                                                                                                              | Minimum number of hours |
|---------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 1.      | Introduction to French Language                                                                                                | 08                      |
| 2.      | Grammar: Articles, Tense, Forms, Numbers, Verbs, Days, Months, Family                                                          | 08                      |
| 3.      | Grammar : Adjectives, Adverbs, Interrogative Forms, Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs | 08                      |
| 4.      | Grammar: Prepositions, Conjunctions, Tenses, Colors, Vegetables, Fruits, Shapes, Verbs                                         | 06                      |

**Total hours (Practical) :30**

**Total hours :30**

**Detail Syllabus:**

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |     |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|
| 1. | Introduction to French Language                                                                                                                                                                                                                                                                                                                                                                                                                                           | 08 Hours | 28% |
|    | <b>Facts and figures about French Language; Basic French Linguistics</b> -* Alphabets * Accents * Liaison * Nasalization<br>French Culture, Differ between French and English; <b>Grammar</b> -Subject Pronoun, Verbs: (être, avoir, habiter, regarder, manger ... “er” verb), Form of address, Numbers (1 to 20), Nouns and plurals of nouns, The expression: C’est, Il y a;<br><b>Presentation:</b> -1) Self-Introduction-2) Question and answering;<br><b>Dialogue</b> |          |     |
| 2. | Grammar: Articles, Tense, Forms, Numbers, Verbs, Days, Months, Family                                                                                                                                                                                                                                                                                                                                                                                                     | 08 Hours | 28% |
|    | <b>Grammar</b> -Definite articles, Indefinite articles, Present tense (Positive Forms, Negative Forms), Numbers (21 to 100, 100-1000), Days, Months, Family, Verbs: (aller, venir, finir, pouvoir, vouloir ... “ir” verb); <b>Social Links</b> -1), My family & relations 2) Appointments 3) Gathering information from someone; <b>Dialogue</b>                                                                                                                          |          |     |
| 3. | Grammar : Adjectives, Adverbs, Interrogative Forms, Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs                                                                                                                                                                                                                                                                                                                                            | 08 Hours | 28% |
|    | <b>Grammar</b> - Common Adjectives, Comparative Adjectives, Common Adverbs, Interrogative Forms, The expression: “On”, Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs: (Prendre, Apprendre, Comprendre, faire ... “re “ verb); <b>Work , Study and Travel</b> -1) Job/ Profession 2) Ticket Reservation (At Bus/At Railway/At Airport); <b>Dialogue</b>                                                                                       |          |     |
| 4. | Grammar: Prepositions, Conjunctions, Tenses, Colours, Vegetables, Fruits, Shapes, Verbs                                                                                                                                                                                                                                                                                                                                                                                   | 06 Hours | 26% |
|    | <b>Grammar</b> -1) Common Prepositions 2) Common Conjunctions 3)Past Tense 4) Future Tense 5) Colors ,Shapes, Animals ,Vegetables, Fruits 6) Verbs: ( “er”, “ir”, “re” etc...); <b>Food &amp; Shopping</b> -1) Buy a vegetables and fruits 2) Any Conversation between Customer and Vendor (At Mall/At Restaurant / At Market); <b>Dialogue</b>                                                                                                                           |          |     |

### Course Outcome (COs):

At the end of the course, the students will be able to

|     |                                                                                                                                |
|-----|--------------------------------------------------------------------------------------------------------------------------------|
| CO1 | Gain basic communication skills in French language with preliminary understanding of grammar                                   |
| CO2 | Develop vocabulary required to speak about him/herself and his/her immediate environment.                                      |
| CO3 | Become capable of interacting in simple ways, to ask simple questions to get necessary information, to reply simple questions. |
| CO4 | Become capable of understanding and using simple instructions in their personal, academic and professional environments.       |
| CO5 | Develop skills and intelligences to function in multi-disciplinary and cross-cultural work environment.                        |
| CO6 | Practice new global trends in communication in multiple perspectives at personal, professional, and social level.              |

### Course Articulation Matrix:

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | -   | -   | -   | -   | -   | -   | -   | -   | 3   | -    | -    | -    | -    | -    |
| CO2 | -   | -   | -   | -   | -   | -   | -   | -   | 3   | -    | -    | -    | -    | -    |
| CO3 | -   | -   | -   | -   | -   | -   | -   | -   | 3   | -    | -    | -    | -    | -    |
| CO4 | -   | -   | -   | -   | -   | -   | -   | -   | 3   | -    | -    | -    | -    | -    |
| CO5 | -   | -   | -   | -   | -   | -   | -   | -   | 3   | -    | -    | -    | -    | -    |
| CO6 | -   | -   | -   | -   | -   | -   | 3   | -   | -   | -    | -    | -    | -    | -    |

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

**Recommended Study Material:**

**Text book:**

1. Complete French: All-In-One, McGraw-Hill, Amazon
2. Best for Grammar: Easy French Step-by-Step, McGraw-Hill, Amazon

**Reference book:**

1. Basic French: McGraw-Hill, Amazon
2. French Grammar for Beginners, Amazon

**Web material:**

1. <https://alison.com/course/french-language-studies-introduction>
2. <https://alison.com/course/basic-french-language-skills-for-everyday-life-revised-2017>
3. <http://www.bbc.co.uk/languages/french/>
4. <https://www.loecsen.com/en/learn-french>
5. <https://www.youtube.com/watch?v=ujDtmohZyII>

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## HSUS501 - ACADEMIC SPEAKING AND PRESENTATION SKILLS

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### Credits and Hours:

| Teaching Scheme | Theory | Practical | Tutorial | Total | Credit    |
|-----------------|--------|-----------|----------|-------|-----------|
| Hours/week      | --     | 30/15     | --       | 30/15 | <b>02</b> |
| Marks           | --     | 100       | --       | 100   |           |

### Pre-requisite courses:

- Beginner/Intermediate level language proficiency

### Outline of the Course:

| Sr. No. | Title of the unit                       | Minimum number of hours |
|---------|-----------------------------------------|-------------------------|
| 1.      | Foundations of Advance Communication    | 04                      |
| 2.      | Art of Conversation                     | 06                      |
| 3.      | Science of Power Speaking               | 06                      |
| 4.      | Academic Speaking Application – Part I  | 08                      |
| 5.      | Academic Speaking Application – Part II | 06                      |

**Total hours (Practical) :30**

**Total hours :30**

**Detail Syllabus:**

|    |                                                                                                                                                                                                                             |          |     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|
| 1. | Foundations of Advance Communication                                                                                                                                                                                        | 04 Hours | 14% |
|    | Meaning and Definition of Advance Communication; Advance Communication in Digital, Social, Mobile World; Strategies for Advance Communication; Meaning and Concept of Academic Language; High Frequency Academic Vocabulary |          |     |
| 2. | Art of Conversation                                                                                                                                                                                                         | 06 Hours | 20% |
|    | Describing people, places and things; Expressing opinions; Making suggestions; Persuading someone; Interpreting and Summarizing                                                                                             |          |     |
| 3. | Science of Power Speaking                                                                                                                                                                                                   | 06 Hours | 20% |
|    | Phonemes, Word Stress, Pronunciation, Intonation, Pause, Register, Fluency, Prosody, Lexical Range                                                                                                                          |          |     |
| 4. | Academic Speaking Application – Part I                                                                                                                                                                                      | 08 Hours | 26% |
|    | Art of Oratory, Formal Presentation, Speech Analysis – Decoding Best Speeches                                                                                                                                               |          |     |
| 5. | Academic Speaking Application – Part II                                                                                                                                                                                     | 06 Hours | 20% |
|    | Job Interview, Group Discussion, Meeting                                                                                                                                                                                    |          |     |

**Course Outcome (COs):**

At the end of the course, the students will be able to

|     |                                                                               |
|-----|-------------------------------------------------------------------------------|
| CO1 | understand and demonstrate advance communication and academic speaking skills |
| CO2 | demonstrate ability to communicate in diverse situations                      |
| CO3 | activate and extend their linguistic and communicative competence             |
| CO4 | demonstrate the formal presentation skills                                    |
| CO5 | demonstrate performing ability at group discussion and personal interview     |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | -   | -   | -   | -   | -   | -   | 2   | -   | 3   | -    | -    | -    | -    | -    |
| CO2 | -   | -   | -   | -   | -   | -   | -   | -   | 3   | -    | -    | -    | -    | -    |
| CO3 | -   | -   | -   | -   | -   | -   | -   | -   | 3   | -    | -    | -    | -    | -    |
| CO4 | -   | -   | -   | -   | -   | -   | -   | -   | 3   | -    | -    | -    | -    | -    |
| CO5 | -   | -   | -   | -   | -   | -   | -   | -   | 3   | -    | 3    | -    | -    | -    |

**Recommended Study Material:**

**Reference book:**

1. *Business Communication Today* (Thirteenth Edition) by Courtland L. Bovee, John V. Thill and Roshan Lal Raina
2. *Effective Speaking Skills* by Terry O' Brien
3. *Speak Better Write Better* by Norman Lewis
4. *Well Spoken: Teaching Speaking to All Students* by Erik Palmer
5. *Let Us Hear Them Speak : Developing Speaking – Listening Skills in English* by Jayshree Mohanraj (Publisher – Sage Publication)
6. *The craft of scientific presentations: Critical steps to succeed and critical errors to avoid.* New York: Springer by Michael Alley
7. *Presentation Skills in English* by Bob Dignen (Publisher: Orient Black Swan)

**Web material:**

1. TED Talk : How to speak so that people want to listen  
[https://www.ted.com/talks/julian\\_treasure\\_how\\_to\\_speak\\_so\\_that\\_people\\_want\\_to\\_listen?language=en](https://www.ted.com/talks/julian_treasure_how_to_speak_so_that_people_want_to_listen?language=en)
2. TED Talk: The 110 techniques of communication and public speaking  
[https://www.ted.com/talks/david\\_jp\\_phillips\\_the\\_110\\_techniques\\_of\\_communication\\_and\\_public\\_speaking](https://www.ted.com/talks/david_jp_phillips_the_110_techniques_of_communication_and_public_speaking)

# **TEACHING SCHEME & DETAILED SYLLABUS**

**FOR**

**M.Sc. (IT) PROGRAMME  
(2<sup>nd</sup> SEMESTER)**

**EFFECTIVE FROM  
ACADEMIC YEAR 2025-26**

**Teaching & Examination Scheme**  
**Master of Science in Information Technology (M. Sc. (IT)) Programme**  
**(Choice Based Credit System)**

**Effective from Year 2025 - 26**  
**Semester – II**

| Course Code                                                                                                                                                                | Course Title                            | Teaching Scheme |           |       |        | Examination Scheme |     |           |     |       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------|-----------|-------|--------|--------------------|-----|-----------|-----|-------|
|                                                                                                                                                                            |                                         | Contact Hours   |           |       | Credit | Theory             |     | Practical |     | Total |
|                                                                                                                                                                            |                                         | Theory          | Practical | Total |        | CCE                | SEE | CCE       | SEE |       |
| CTUE509-CTUE512                                                                                                                                                            | Elective-II                             | 3               | 3         | 6     | 6      | 50                 | 50  | 50        | 50  | 200   |
| CTUC513                                                                                                                                                                    | Web Development Using Open Source       | 3               | 3         | 6     | 6      | 50                 | 50  | 50        | 50  | 200   |
| CTUC514                                                                                                                                                                    | Advanced Mobile Application Development | 3               | 3         | 6     | 6      | 50                 | 50  | 50        | 50  | 200   |
| CTUC515                                                                                                                                                                    | Full Stack Web Development              | 3               | 3         | 6     | 6      | 50                 | 50  | 50        | 50  | 200   |
| HSUA501                                                                                                                                                                    | Academic Writing                        | -               | 2         | 2     | 2      | -<br>-             | -   | 25        | 25  | 50    |
|                                                                                                                                                                            | University Elective-I **                | -               | 2         | 2     | 2      | -                  | -   | 50        | 50  | 100   |
|                                                                                                                                                                            |                                         | 12              | 16        | 28    | 28     | 400                |     | 550       |     | 950   |
| *CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation |                                         |                 |           |       |        |                    |     |           |     |       |

\*\*Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **CAUD516-Internet and Web Designing** and **CAUD517-Mobile Application Development** course for others.

| <b>Elective-II</b> |                                             |
|--------------------|---------------------------------------------|
| <b>Course Code</b> | <b>Course Title</b>                         |
| CTUE509            | Game Design                                 |
| CTUE510            | Advanced Python Programming                 |
| CTUE511            | Blockchain Programming                      |
| CTUE512            | HTTP Web Service for Enterprise Application |

### University Elective-I

| <b>Sr. No.</b> | <b>Course Code</b> | <b>Course Name</b>                                                   | <b>Department/Faculty</b> |
|----------------|--------------------|----------------------------------------------------------------------|---------------------------|
| 1              | FTUD552            | Health and Safety Management                                         | Engineering               |
| 2              | FTUD502            | Sustainable Engineering Concepts and Life Cycle Analysis             | Engineering               |
| 3              | FTUD501            | Blockchain and its Applications                                      | Engineering               |
| 4              | OCMPH1003          | Introduction On Intellectual Property To Engineers And Technologists | Pharmacy                  |
| 5              | MBUD558            | Introduction to Operations Research                                  | Management                |
| 6              | CAUD518            | Software Project Management                                          | Computer Science          |
| 7              | FSUD551            | Energy resources, economics and sustainability                       | Applied Science           |
| 8              | FSUD552            | Nuclear Astrophysics                                                 | Applied Science           |
| 9              | FSUD553            | Computational Science in Engineering                                 | Applied Science           |
| 10             | FSUD554            | Plastic Waste Management                                             | Applied Science           |
| 11             | PTUD796            | Yoga and Positive Psychology for Managing Career and Life            | Physiotherapy             |
| 12             | NRMD551            | Mindfulness and Well-being: Living with Balance and Ease             | Nursing                   |

## CTUE509 - Game Design

### (200 Marks)

**Contact Hours: 06**

**Pre-requisite:** Knowledge of C++, physics and mathematics concepts

**Methodology & Pedagogy:** During theory lectures illustrations Graphics, animation and various concepts regarding Game Development. Emphasize will be given on some mathematical and physics concepts, Fundamental of Graphics and Graphics Programming, Vectors, Image Data, 2D graphics, Collision detections, Fundamental of Game programming, Gameloop, Game Engine and many more. During Practical sessions, students will be developing Simple 2D graphics, Implement animation on that graphics, Handles Input and Output of data according to event generation. Student will develop simple 2D game using Visual C++.

**Outline of the course:**

| Unit Number | Title of the Unit                                  | Minimum Number of Hours |           |
|-------------|----------------------------------------------------|-------------------------|-----------|
|             |                                                    | Theory                  | Practical |
| 1           | Basics of Graphics and Game                        | 04                      | 36        |
| 2           | Important Aspects of Mathematics and Physics       | 07                      |           |
| 3           | Graphics Programming Fundamentals                  | 05                      |           |
| 4           | Animation, Transformations and Collision Detection | 07                      |           |
| 5           | Actual Game Programming                            | 09                      |           |
| 6           | Game Engine Introduction                           | 04                      |           |

**Total Hours (Theory): 36**

**Total Hours (Lab): 36**

**Total Hours: 72**

## **Detail Syllabus:**

### **Unit I: Basics of Graphics and Game**

**Hours 04**

The World of Computer Graphics, Current and Future Application Areas, User-Interface Considerations, History of Games and Game Development.

### **Unit II: Important Aspects of Mathematics and Physics**

**Hours 07**

Vector Addition, Subtraction, Dot Product, Cross Product, Magnitude, Finding the Normal of a Vector, Projection of one vector over another vector, Matrix and operations: Matrix Addition, Subtraction, Multiplication, Transpose, and Inverse of a Matrix, Basic Trigonometry: Meaning of Sin, Cosine, and Tan functions, Cartesian coordinate system, Polar Coordinate System, Nested Coordinate Systems Physics Concepts: Speed, Velocity, Acceleration, Mass, Force, Gravity, Friction.

### **Unit III: Graphics Programming Fundamentals**

**Hours 05**

Video Ram, Pixels and screen sizes, Graphics Card, Frame Buffer, Double Buffer, Screen, Refresh Rates (on LCD display units), Screen Resolutions, Vector Graphics and Raster Graphics, Image Data - How are images represented in a Bitmap file, Drawing images on a screen.

### **Unit IV: Animation, Transformations and Collision Detection**

**Hours 07**

Programming Simple Animations and understanding animation frame rates, Concept of Sprites in Animation, Line Drawing Algorithm, Circle Drawing Algorithm, Transformations - Translate, Rotate, Scale, Transformation Matrix, Collision detection techniques for 2D Shapes: Circle to Circle, Rectangle to Rectangle, Circle to Line, Circle to Rectangle, Line to Line.

### **Unit V: Actual Game Programming**

**Hours 09**

Application and Window Management, Input Management, File Input and Output, Audio, Programming, Game loop, Frame rate independent entity behaviors.

### **Unit VI: Game Engine Introduction**

**Hours 04**

Introduction to Game Engine, Types of Game Engines, Some of the most popular Game Engines of today: Unity, Unreal, Cocos2d-X, Box2d.

## **Core Books:**

1. Laszlo Michael, Computational Geometry and Computer Graphics in C++, Pearson Education; First edition, 2017.
2. Roger Mayne, Introduction to Windows and Graphics Programming with Visual C++.NET, World Scientific Publishing Co. Pte. Ltd., 2005.

3. John Horton, Beginning C++ Game Programming, Packt Publishing Limited, 2016.
4. Michael Dawson, Beginning C++ Through Game Programming, Cengage; 4th edition, 2014.
5. Eric Lengyel, Foundations of Game Engine Development, Volume 1: Mathematics, Terathon Software LLC; 1 edition, 2017.

#### Reference Books:

1. John F. Hughes, Andries van Dam, Morgan McGuire, David F. Sklar, James D. Foley, Steven K. Feiner, Kurt Akeley, "Computer Graphics: Principles and Practice in C", 3rd Edition, Addison Wesley Professional, 2013
2. David Conger, Ron Little, Creating Games in C++: A Step-by-step Guide, Pearson Education, 2006.
3. Fabien Sanglard, Game Engine Black Book: Wolfenstein 3D, Create Space Independent Publishing Platform; 1 edition
4. Constantine Pozrikidis, Introduction to C++ Programming and Graphics, Springer-Verlag New York Inc.; 2007
5. James M. Van Verth, Lars M. Bishop, Essential Mathematics for Games and Interactive Applications, A K Peters/CRC Press; 3rd edition, 2015

#### Web References:

1. <http://cplusplus.happycodings.com/computer-graphics/> [Graphics Tutorial]
2. <http://cplusplus.happycodings.com/games/> [Game Tutorial]
3. <https://www.geeksforgeeks.org/basic-graphic-programming-in-c/>
4. <https://www.thecrazyprogrammer.com/2013/04/simple-program-to-create-moving-car-in.html>
5. <http://mytechnotrick.blogspot.com/2015/07/#>
6. <https://www.youtube.com/watch?v=hfBR6ULOOjw>

**Course Outcomes:** Upon successful completion of the course, students will be,

|       |                                                                                  |
|-------|----------------------------------------------------------------------------------|
| CO1 : | Be having basic knowledge of computer graphics                                   |
| CO2 : | Be familiar with tools and techniques used for graphics and game programming     |
| CO3 : | Be able to perform mathematic calculation required for game programming          |
| CO4 : | Be familiar with the various image data and shapes.                              |
| CO5 : | Be able to create animated and interactive graphics which lead to develop a game |

**Course Outcomes Mapping:**

| Unit No. | Unit Name                                          | Course Outcomes |     |     |     |     |
|----------|----------------------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                                    | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | Basics of Graphics and Game                        | √               |     |     |     |     |
| 2        | Important Aspects of Mathematics and Physics       |                 | √   |     |     |     |
| 3        | Graphics Programming Fundamentals                  | √               |     | √   |     |     |
| 4        | Animation, Transformations and Collision Detection |                 |     | √   |     |     |
| 5        | Actual Game Programming                            |                 |     |     | √   |     |
| 6        | Game Engine Introduction                           |                 |     |     |     | √   |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 3   | 2   | -   | 1   | 2   | -   | 2   | 1   | 1   | -    | 1    | 1    | 2    | 3    |
| CO2 | 3   | 2   | -   | 2   | 2   | -   | 2   | 1   | 1   | -    | 1    | 1    | 2    | 1    |
| CO3 | 3   | 1   | 1   | 3   | 2   | -   | 2   | 1   | 1   | -    | 2    | 2    | 2    | 1    |
| CO4 | 3   | 2   | 2   | 2   | 2   | 1   | 2   | 3   | 1   | 2    | 3    | 3    | 3    | 2    |
| CO5 | 3   | 3   | 3   | 2   | 2   | 1   | 2   | 3   | 1   | 3    | 3    | 3    | 3    | 3    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## CTUE510 - Advanced Python Programming (200 Marks)

**Contact Hours: 06**

**Pre-requisite:** Basics of Programming.

**Methodology & Pedagogy:** Theory sessions are required to address computational power of python through its ability to deploy programs using functional, object oriented and advance python based aspects. Practical sessions demonstrate the implementation of the concepts which are taught during the theory sessions. Case study will help the students to come out with one working module in any of the advanced python programming.

**Outline of the course:**

| Unit Number | Title of the Unit                             | Minimum Number of Hours |           |
|-------------|-----------------------------------------------|-------------------------|-----------|
|             |                                               | Theory                  | Practical |
| 1           | Fundamentals of Python                        | 6                       | 36        |
| 2           | Iterable Object in Python and Function        | 7                       |           |
| 3           | Object oriented Programming with Python       | 5                       |           |
| 4           | GUI Programming using Python                  | 6                       |           |
| 5           | Python libraries and Data collection          | 6                       |           |
| 6           | Use of Python to develop AI based application | 6                       |           |

**Total Hours (Theory): 36**

**Total Hours (Lab): 36**

**Total Hours: 72**

## **Detail Syllabus:**

### **Unit I: Fundamentals of Python**

**Hours 06**

Introduction to Python Program, Features of Python. Environmental setup for Python Programming, Basic program structure to write Python, Data Type, Input and Output Statements, Comment in Python, If Statement with its variation, While and For Loop.

### **Unit II: Iterable Object in Python and Function**

**Hours 07**

List, Tuple, Set, Dictionary and custom iterable object What are functions, calling functions, creating functions, passing functions, formal arguments, variable length arguments, default arguments, returning values from the functions, returning multiple values from the functions.

### **Unit III: Object oriented Programming with Python**

**Hours 05**

Basics of class, object and instance. Class level attribute and instance level attribute. Constructor and other magic methods. Bound and unbound methods. Built in functions for python class and objects

### **Unit IV: GUI Programming using Python**

**Hours 06**

Introduction, Python's GUI library: Tkinter (installing and getting started), Introduction to GUI programming, top-level window, Tk widgets: Label, Button, scale, scroll bar, text, Checkbutton, Radiobutton. Developing GUI based applications using Tkinter.

### **Unit V: Python libraries and Data collection**

**Hours 06**

NumPy: Introduction, 2D arrays, Numpy arraya Vs. Python List. Pandas: Introduction, importing data with pandas, Data frames and Series, pre-processing phase of data analysis, handling missing values in data, formatting data to standardize it and make it consistent, normalizing data

### **Unit VI: Use of Python to develop AI based application**

**Hours 06**

ML: Introduction to machine learning- Supervised and Unsupervised Learning, Model development using Linear Regression, Model Visualization, Prediction and Decision Making, Model Evaluation: Over-fitting, Under-fitting and Model Selection  
DIP: Introduction to digital image Processing, Basic function to import draw and display images using Python.

**Core Books:**

1. Wesley J. Chun : Core Python Programming, 2nd edition, Pentice Hall, 2006.
2. Magnus Lie Hetland : Beginning Python from novice to professional, 2nd edition, Apress, 2009.

**Reference Books:**

1. Mark Lutz : Programming Python, 4th Edition, O'reilly, 2011 .
2. Dusty Philips: Python 3 Object oriented Programming, PACKT publishing, 2010.
3. Steve Holden: Python Web Programming, 1st edition, 2002.

**Web References:**

1. <https://developers.google.com/edu/python/> [Google's Python class]
2. <https://www.learnpython.org/> [Learnpython.org]
3. <https://docs.python.org/3/tutorial/> [Python tutorial]
4. <https://www.tutorialspoint.com/python/index.htm> [Tutorials point]

**Course Outcomes:** Upon successful completion of the course, students will be,

|       |                                                                                                        |
|-------|--------------------------------------------------------------------------------------------------------|
| CO1 : | Students will be able to understand scripting and programming constructs of Python.                    |
| CO2 : | Understanding Python's iterable objects and functions.                                                 |
| CO3 : | Clear Understanding of the object-oriented concepts using Python.                                      |
| CO4 : | Students exposed to GUI programming and Python libraries to handle data.                               |
| CO5 : | Students will learn model development using advance concepts of Machine Learning and its applications. |

**Course Outcomes Mapping:**

| Unit No. | Unit Name                                     | Course Outcomes |     |     |     |     |
|----------|-----------------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                               | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | Fundamentals of Python                        | √               |     |     |     |     |
| 2        | Iterable Object in Python and Function        |                 | √   |     |     |     |
| 3        | Object oriented Programming with Python       |                 |     | √   |     |     |
| 4        | GUI Programming using Python                  |                 |     |     | √   |     |
| 5        | Python libraries and Data collection          |                 |     |     | √   |     |
| 6        | Use of Python to develop AI based application |                 |     |     |     | √   |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 2   | 2   | 3   | 2   | 2   | 2   | 3   | 2   | 1   | 1    | 1    | 1    | 3    | 2    |
| CO2 | 2   | 3   | 3   | 3   | 3   | 2   | 3   | 2   | 1   | 1    | 1    | 1    | 3    | 2    |
| CO3 | 2   | 3   | 3   | 3   | 3   | 2   | 3   | 3   | 2   | 1    | 2    | 1    | 3    | 2    |
| CO4 | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 2   | 2    | 3    | 2    | 3    | 2    |
| CO5 | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 1   | 2    | 3    | 3    | 3    | 3    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## **CTUE511 – Blockchain Programming (200 Marks)**

**Contact Hours: 06**

**Pre-requisite: None**

**Methodology & Pedagogy:** During the lecture sessions, the students will learn about various sub systems that work in integrated manner to make blockchain work. The teacher will introduce students to concepts of bitcoin and ethereum technologies. The students will also understand about application domains where blockchain can be applied. In the lab sessions, the students will learn to program smart contracts with solidity on ethereum platform.

**Outline of the course:**

| Unit Number | Title of the Unit                         | Minimum Number of Hours |           |
|-------------|-------------------------------------------|-------------------------|-----------|
|             |                                           | Theory                  | Practical |
| 1           | Introduction to blockchain                | 07                      | 36        |
| 2           | Functioning of blockchain                 | 07                      |           |
| 3           | Bitcoin and related concepts              | 05                      |           |
| 4           | Ethereum and related concepts             | 09                      |           |
| 5           | Introduction to solidity programming      | 04                      |           |
| 6           | Advanced concepts in solidity programming | 04                      |           |

**Total Hours (Theory): 36**

**Total Hours (Lab): 36**

**Total Hours: 72**

**Detail Syllabus:**

**Unit I: Introduction to blockchain**

**Hours 07**

History of blockchain, Centralized, decentralized and distributed systems, Layers of blockchain, Importance of blockchain, Applications of blockchain

**Unit II: Functioning of blockchain**

**Hours 07**

Cryptographic concepts- Symmetric key cryptography, Cryptographic hash function, Asymmetric key cryptography, Byzantine general's problem, Merkel tree, Structure of block

**Unit III: Bitcoin and related concepts**

**Hours 05**

What is bitcoin? Working with bitcoin, Bitcoin network, Bitcoin mining, Consensus algorithms

**Unit IV: Ethereum and related concepts**

**Hours 09**

Design of ethereum, Ethereum blockchain, Smart contracts in ethereum, Ethereum virtual machine, Ethereum eco system

**Unit V: Introduction to solidity programming**

**Hours 04**

Ethereum networks, Geth, Solidity compiler, Web3 library, Metamask wallet, emix, Ganache, Structure of ethereum smart contract, datatypes in solidity

**Unit VI: Advanced concepts in solidity programming**

**Hours 04**

Global variables and functions, expressions and control structures, writing smart contracts in solidity

**Core Books:**

1. Bikramaditya Singhal, Gautam Dhameja, Priyansu Sekhar Panda: Beginning Blockchain - A Beginner's Guide to Building Blockchain Solutions: 1st Edition: APress Publication: 2018.
2. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, Steven Goldfeder: Bitcoin and Cryptocurrency Technologies - A Comprehensive Introduction: 1st Edition: Princeton University Press: 2016.
3. Ritesh Modi: Solidity Programming Essentials - A Beginner's Guide to Build Smart Contracts for Ethereum and Blockchain: 1st Edition: Packt Publishing: 2018

**Reference Books:**

1. Fabian Schar, Aleksander Berentsen: Bitcoin, Blockchain, and Cryptoassets: 1st Edition: MIT Press: 2020
2. Kevin Solorio, Randall Kanna, David H. Hoover: Hands-On Smart Contract Development with Solidity and Ethereum From Fundamentals to Deployment: 1<sup>st</sup> Edition: O'Reilly Media: 2019

**Web References:**

1. <https://andersbrownworth.com/cms/460/blockchain/demo> [Blockchain Demo]
2. <https://ethdocs.org/en/latest/introduction/what-is-ethereum.html> [Ethereum Introduction]
3. <https://www.youtube.com/watch?v=M576WGiDBdQ> [Solidity Programming]

**Course Outcomes:** Upon successful completion of the course, students will be,

|       |                                                                                                                        |
|-------|------------------------------------------------------------------------------------------------------------------------|
| CO1 : | The students will learn about underlying technologies of blockchain and various applications of blockchain technology. |
| CO2 : | The students will understand working of blockchain in detail.                                                          |
| CO3 : | The students will learn about how the blockchain is used to power bitcoin.                                             |
| CO4 : | The students will see how ethereum utilizes blockchain.                                                                |
| CO5 : | The students will be able to create and deploy solidity based smart contracts on ethereum platform.                    |

**Course Outcomes Mapping:**

| Unit No. | Unit Name                                 | Course Outcomes |     |     |     |     |
|----------|-------------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                           | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | Introduction to blockchain                | √               |     |     |     |     |
| 2        | Functioning of blockchain                 |                 | √   |     |     |     |
| 3        | Bitcoin and related concepts              |                 |     | √   |     |     |
| 4        | Ethereum and related concepts             |                 |     |     | √   |     |
| 5        | Introduction to solidity programming      |                 |     |     |     | √   |
| 6        | Advanced concepts in solidity programming |                 |     |     |     | √   |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 2   | 2   | 2   | 1   | 3   | 1   | 2   | 1   | 2   | -    | -    | 1    | 1    | -    |
| CO2 | 3   | -   | 2   | 2   | 2   | 1   | 1   | 2   | 1   | -    | -    | 1    | 2    | -    |
| CO3 | 2   | 2   | 3   | 3   | 3   | 2   | 2   | 1   | 2   | -    | -    | 1    | 2    | -    |
| CO4 | 3   | -   | 3   | 3   | 3   | 2   | 3   | 3   | 1   | -    | 1    | 2    | 3    | -    |
| CO5 | 3   | -   | 3   | 3   | 3   | 2   | 3   | 3   | 2   | -    | 2    | 3    | 3    | -    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## **CTUE512 - HTTP Web Service for Enterprise Application**

### **(200 Marks)**

**Contact Hours: 06**

**Pre-requisite:** Work experience with C#, ASP.NET MVC, MSSQL, HTML, CSS, JavaScript and have some understanding of JQuery.

**Methodology & Pedagogy:** During theory lectures, illustrations emphasizing the need for advance features of WEB API and ASP.NET will be covered. During Practical sessions, students will require to develop Web API using concepts of .NET framework and other pre-requisite technologies, discussed during class.

**Outline of the course:**

| Unit Number | Title of the Unit                                | Minimum Number of Hours |           |
|-------------|--------------------------------------------------|-------------------------|-----------|
|             |                                                  | Theory                  | Practical |
| 1           | ASP.NET Web API - Overview                       | 4                       | 36        |
| 2           | MVC and Web API Controller with Entity Framework | 8                       |           |
| 3           | Web API Routing                                  | 4                       |           |
| 4           | Web API (Action Formatter and Filters)           | 4                       |           |
| 5           | Backend Validation with Exception Handling       | 8                       |           |
| 6           | AJAX and JQuery                                  | 8                       |           |

**Total Hours (Theory): 36**

**Total Hours (Lab): 36**

**Total Hours: 72**

## **Detail Syllabus:**

### **Unit I: ASP.NET Web API - Overview**

**Hours 04**

Introduction to RESTful WEB API, Characteristics of ASP.NET WEB API, versions of Web API, Difference among web service, Window Communication Foundation and WEB API, Test Web API Fiddler and Postman

### **Unit II: MVC and Web API Controller with Entity Framework**

**Hours 08**

Use of Web API Controller in Controller class of Web API, Functionality of Web API Controller, Difference between Web API controller and MVC controller, Action Method Naming Conventions, Action Result, MVC with WEB API, CRUD operation with Entity Framework in Web API.

### **Unit III: Web API Routing**

**Hours 04**

Routing in Web API, Routing: Convention-based Routing and Attribute based Routing, Routing and Action Execution.

### **Unit IV: Web API (Action Formatter and Filters)**

**Hours 04**

Data Formatter, Media Type Formatter, Web API Filter, Exception Filters: HttpResponseMessage, Exception Filters, Registering Exception Filters, HttpError

### **Unit V: Backend Validation with Exception Handling**

**Hours 08**

Model Validation with annotation, Custom Exception, recognizing need of custom Exceptions, Backend validation using Custom Exception for robustness and Data Integrity.

### **Unit VI: AJAX and JQuery**

**Hours 08**

Consume RESTful Web API through GET, POST, PUT and DELETE. Understanding JQuery and JavaScript. JQuery usage and AJAX Request configuration and parameters understanding.

## **Core Books:**

1. Mithun Pattankar, Malendra Hurbuns, Mastering ASP.NET Web API: Build powerful HTTP services and make the most of the ASP.NET Core Web API platform 1st Edition. – 2017.

## **Reference Books:**

1. Jamie Kurtz, Brian Wortman, ASP.NET Web API 2: Building a REST Service from Start to Finish. - 2014

### Web References:

1. [https://www.tutorialspoint.com/asp.net\\_mvc/asp.net\\_mvc\\_web\\_api.htm](https://www.tutorialspoint.com/asp.net_mvc/asp.net_mvc_web_api.htm)

**Course Outcomes:** Upon successful completion of the course, students will be,

|       |                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------|
| CO1 : | Be familiar with RESTful API and ASP .NET Web API.                                                            |
| CO2 : | Be able to implement ASP .NET Web API Controller and configure its Routing and implementing Entity Framework. |
| CO3 : | Be able to Configure actions in API Controllers, Filters and Formatters.                                      |
| CO4 : | Be able provide Data Integrity with validations with Exceptions.                                              |
| CO5 : | Be able to call API through testing applications and AJAX calls from 3 <sup>rd</sup> Party Applications.      |

### Course Outcomes Mapping:

| Unit No. | Unit Name                                        | Course Outcomes |     |     |     |     |
|----------|--------------------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                                  | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | ASP.NET Web API - Overview                       | √               |     |     |     |     |
| 2        | MVC and Web API Controller with Entity Framework |                 | √   |     |     |     |
| 3        | Web API – Routing                                |                 | √   |     |     |     |
| 4        | Web API (Action Formatter and Filters)           |                 |     | √   |     |     |
| 5        | Backend Validation with Exception Handling       |                 |     |     | √   |     |
| 6        | AJAX and JQuery                                  |                 |     |     |     | √   |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 1   | 3   | 3   | 1   | 3   | -   | -   | 2   | 2   | 2    | 2    | 2    | 2    | 2    |
| CO2 | 2   | 1   | 2   | 2   | 3   | 2   | 2   | 3   | -   | 3    | 2    | 2    | 3    | 3    |
| CO3 | 2   | 2   | 2   | 2   | 3   | 2   | 2   | 2   | -   | 1    | 2    | 2    | 3    | 2    |
| CO4 | 3   | 3   | 3   | 3   | 2   | 3   | 2   | 2   | -   | 2    | 3    | 2    | 3    | 2    |
| CO5 | 2   | 3   | 3   | 1   | 3   | 2   | 2   | 3   | 2   | 2    | 3    | 2    | 3    | 2    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## CTUC513 - Web Development using Open Source (200 Marks)

**Contact Hours: 06**

**Pre-requisite:** Basic understanding of HTML and MySQL.

**Methodology & Pedagogy:** During theory and practical sessions, students able to install & configure PHP and prerequisite software(s). Also, student will be emphasized to develop dynamic web applications.

**Outline of the course:**

| Unit No. | Title of the Unit                                             | Minimum Number of Hours |           |
|----------|---------------------------------------------------------------|-------------------------|-----------|
|          |                                                               | Theory                  | Practical |
| 1        | Basics of HTML and Java Script                                | 06                      | 36        |
| 2        | Introduction to Open Source Software and PHP, Basics of PHP   | 05                      |           |
| 3        | Control Structure, Function, Array and Overview of OOP in PHP | 06                      |           |
| 4        | Form Handling Using PHP                                       | 07                      |           |
| 5        | PHP Utilities                                                 | 05                      |           |
| 6        | Relational Database Using PHP                                 | 07                      |           |

**Total Hours (Theory): 36**

**Total Hours (Lab): 36**

**Total Hours: 72**

**Detail Syllabus:**

**Unit I: Basics of HTML and Java Script**

**Hours: 06**

Basics of HTML, Introduction of JavaScript, Variables, Overview of operators, Control statements and looping statement, Overview of DOM, function declaration and calling with event.

**Unit II: Introduction to Open Source Software and PHP, Basics of PHP**

**Hours: 05**

Overview of Open Source Software, Installation & Configuration of PHP, Introduction to PHP, Working of HTML with PHP, PHP language Basics: Lexical Structure, Data types, Variables, Expressions and Operators.

**Unit III: Control Structure, Function, Array and Overview of OOP in PHP**

**Hours: 06**

PHP language Basics: Control and Looping statements. Functions: Function Definition, Function Parameters, Returning Values. Strings: Usages and String Functions, Arrays: Types of Arrays and its Usages, Array functions. Overview: Objects, Declaring Class, Properties, Methods, Exception Handling, Examples.

**Unit IV: Form Handling Using PHP**

**Hours: 07**

Capturing data with PHP Using HTML Form Elements, Send Form data using GET Method & POST Method, Receive Form data using \$\_GET, \$\_POST & \$\_REQUEST variables, Super Global Variables, State Management Techniques: Concept of Session, starting session, modifying session variables, Un registering and deleting session variable, Concept of Cookies.

**Unit V: PHP Utilities**

**Hours: 05**

File Uploading: Upload Single and Multiple file using PHP script, Understanding HTTP requests, Exploring and modifying HTTP responses, getting information from web server.

**Unit VI: Relational Databases Using PHP**

**Hours: 07**

Relational Databases and SQL, Using PHP to access Databases, CRUD operations, Handling Errors.

**Core Books:**

1. Vikram Vaswani, PHP: A Beginner's Guide: Indian Edition, First Edition, McGraw Hill, 2009
2. Matt Doyle , Beginning PHP 5.3 , Wrox , 2010
3. Ballard and Moncur, Teach Yourself Javascript in 24 Hours, Sams Publishing, 2015

### Reference Books:

1. Timothy Boronczyk , Elizabeth Naramore, Jason Gerner, Yann Le Scouarnec, Jeremy Stolz, Michael K. Glass): Beginning PHP6, Apache, MySQL Web Development: Wrox , 2009.
2. Lynn Beighley and Michael Morrison, Head First PHP & MySQL, First Edition, O'Reilly Publication, 2009.

### Web References:

1. <https://github.com/PHPMailer/PHPMailer> [PHP Mailer Code]
2. <https://www.php.net/> [Official website of PHP]
3. <https://www.geeksforgeeks.org/php-tutorials/> [Lecture notes of PHP]
4. <https://www.w3schools.com/php/default.asp> [Lecture notes of PHP]

**Course Outcomes:** Upon successful completion of the course, students will be,

|       |                                                                                                        |
|-------|--------------------------------------------------------------------------------------------------------|
| CO1 : | Gain understanding of HTML and Java Script.                                                            |
| CO2 : | To utilize knowledge and skills for basics of PHP and functions of PHP                                 |
| CO3 : | Acquire the knowledge of array and OOP in PHP                                                          |
| CO4 : | Learn Form handling and utilities in PHP.                                                              |
| CO5 : | Be able to develop dynamic web based application using PHP and MySQL with state management techniques. |

### Course Outcomes Mapping: -

| Unit No. | Unit Name                                                     | Course Outcomes |     |     |     |     |
|----------|---------------------------------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                                               | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | Basics of HTML and Java Script                                | √               |     |     |     |     |
| 2        | Introduction to Open Source Software and PHP, Basics of PHP   |                 | √   |     |     |     |
| 3        | Control Structure, Function, Array and Overview of OOP in PHP |                 |     | √   |     |     |
| 4        | Form Handling Using PHP                                       |                 |     |     | √   |     |
| 5        | PHP Utilities                                                 |                 |     |     | √   |     |
| 6        | Relational Database Using PHP                                 |                 |     |     |     | √   |

**Course Articulation Matrix: -**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 3   | 3   | 3   | 3   | -   | 1   | -   | 1   | 1   | -    | -    | 3    | -    | 3    |
| CO2 | 3   | 3   | 3   | 3   | -   | 1   | -   | 1   | 1   | -    | -    | 3    | -    | 3    |
| CO3 | 3   | 3   | 2   | 3   | -   | 1   | -   | 1   | 2   | -    | -    | 2    | -    | 3    |
| CO4 | 3   | 3   | 3   | 3   | -   | 1   | -   | 1   | 1   | -    | -    | 3    | -    | 3    |
| CO5 | 3   | 3   | 3   | 3   | 2   | 1   | 1   | 1   | 2   | 1    | 1    | 3    | 3    | 3    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## CTUC514 - Advanced Mobile Application Development (200 Marks)

**Contact Hours: 06**

**Pre-requisite:** Object Oriented Programming Concepts

**Methodology & Pedagogy:** During theory lectures illustrations emphasizing the need for basic features of Mobile Computing and Cross Platform- the Mobile Application Development platform will be given. During Practical sessions, students will be required to develop Mobile Application using Dart language in Flutter. Student shall also develop applications with elegant user interface that deal with data storage using Firebase and state management

**Outline of the course:**

| Unit Number | Title of the Unit                                          | Minimum Number of Hours |           |
|-------------|------------------------------------------------------------|-------------------------|-----------|
|             |                                                            | Theory                  | Practical |
| 1           | Introduction to Cross-platform advance with Flutter & Dart | 05                      | 36        |
| 2           | UI Designing with Flutter                                  | 07                      |           |
| 3           | State management with App creation                         | 05                      |           |
| 4           | Architecting Flutter applications and Its packages         | 08                      |           |
| 5           | Introduction to Backend to Flutter-I                       | 05                      |           |
| 6           | Introduction to Backend to Flutter-II                      | 06                      |           |

**Total Hours (Theory): 36**

**Total Hours (Lab): 36**

**Total Hours: 72**

**Detail Syllabus:**

**Unit I: Introduction to Cross-platform advance with Flutter & Dart Hours 05**

Set up a new Flutter project using Android Studio. Widget tree, Interface design: pre-made Flutter Widgets. Image and Text Widgets. App Icons for iOS and Android. Add and load image assets to Flutter projects. Run Flutter apps on iOS Simulator, Android Emulator and physical devices.

**Unit II: UI Designing with Flutter**

**Hours 07**

Hot Reload and Hot Restart, Use of Pubspec.yaml file, custom assets and fonts. An introduction to the Widget build(), layout widgets: Columns, Rows, Containers and Cards. Material icons, Icons class. Theme widgets. Refactoring widgets. Dart annotations and modifiers. Immutability of Stateless and Stateful Widgets. Update screen with the build(). Custom Flutter Widgets. Difference between final and const in Dart. Maps, enums and the ternary operator. Functions and arguments in Dart. Multi-screen Flutter apps, routes and the Navigator widget. Flutter favours composition vs. inheritance (customizing widgets).

**Unit III: State management with App creation**

**Hours 05**

About Stateful and Stateless Widgets, callbacks. Declarative style of UI programming, Flutter widgets react to state changes. Import Dart libraries. Variables, data types and functions in Dart. Build flexible layouts. Understand the relationship between setState(), State objects and Stateful Widgets.

**Unit IV: Architecting Flutter applications and Its packages**

**Hours 08**

Dart package manager, Flutter compatible packages. The structure of the pubspec.yaml file. Incorporate the audio players package to play sound. Functions in Dart and arrow syntax. Refactor widgets, Flutter's philosophy of UI as code. The lists and conditionals in Dart. Classes and objects. Understand Object Oriented Dart. Dart Constructors for Flutter widgets. Design patterns to structure Flutter apps. Structuring and organizing Flutter apps.

**Unit V: Introduction to Backend to Flutter-I**

**Hours 05**

Asynchronous programming in Dart and use of async/await. Stateful Widget lifecycle methods, Handling exceptions in dart. Null aware operators. Location data from both iOS and Android. Http package and live data from open APIs. Parse JSON data using the dart:convert library.

**Unit VI: Firebase and State Management with Flutter-II**

**Hours 06**

State objects via the Stateful Widget. Use the TextField Widget to take user input. Pass data backwards using the Navigator widget. Hero animations in Flutter apps. Animation

controller, custom animations. The use of Dart mixins. Firebase Cloud Firestore into your Flutter apps. Authentication with Firebase Auth package.

**Core Books:**

1. Marco L. Napoli: Beginning Flutter: A Hands On Guide to App Development: Wrox publication: 2019.

**Reference Books:**

1. Eric Windmill: Flutter in Action: Edition: Manning Publication: January 2020.
2. Alessandro Biessek: Flutter for Beginners: An introductory guide to building cross-platform mobile applications with Flutter and Dart 2: Packt publication: September 2019

**Web References:**

1. <https://docs.flutter.dev/reference/tutorials>
2. <https://www.tutorialspoint.com/flutter/index.htm>
3. <https://www.javatpoint.com/flutter>
4. <https://fluttertutorial.in/>

**Course Outcomes:** Upon successful completion of the course, students will be,

|       |                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------|
| CO1 : | able to clear all object oriented programming and cross platform concepts                                            |
| CO2 : | able to learn Flutter and Dart step by step                                                                          |
| CO3 : | able to learn the reduce the code through native app performance, animated UI with material design and least testing |
| CO4 : | able to use Firebase to authenticate the users and use the remote database                                           |
| CO5 : | able to build engaging native mobile apps for both Android and iOS                                                   |

**Course Outcomes Mapping:**

| Unit No. | Unit Name                                                          | Course Outcomes |     |     |     |     |
|----------|--------------------------------------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                                                    | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | Introduction to cross-platform development with Flutter and Dart   | √               |     |     |     |     |
| 2        | Creating UI with Flutter                                           | √               | √   |     |     |     |
| 3        | Building Apps with state                                           |                 |     | √   |     |     |
| 4        | Leveraging Flutter packages and structuring Flutter applications   |                 |     | √   |     |     |
| 5        | Incorporating backend data with Flutter application                |                 |     |     |     | √   |
| 6        | Integrating Flutter Application with Firebase and State Management |                 |     |     | √   | √   |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 3   | -   | -   | 3   | 2   | 1   | 3   | -   | 1   | -    | 1    | -    | -    | 1    |
| CO2 | 3   | 1   | 3   | 2   | 3   | 1   | 2   | 1   | 1   | -    | 1    | 1    | 1    | 2    |
| CO3 | 3   | -   | 3   | 1   | 3   | 1   | 1   | 2   | 1   | -    | 1    | 1    | 1    | 2    |
| CO4 | 3   | 3   | 3   | 3   | 3   | 1   | 1   | 3   | 1   | -    | 2    | 2    | 3    | 1    |
| CO5 | 3   | 3   | 3   | 1   | 3   | 1   | 1   | 3   | 1   | 1    | 2    | 2    | 3    | 2    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## CTUC515 - Full-Stack Web Development (200 Marks)

**Contact Hours: 06**

**Pre-requisite:** Working knowledge of HTML and JavaScript

**Methodology & Pedagogy:** During the theory sessions students will understand the various JavaScript frameworks and its architecture, also able to comprehend MEAN stack application concepts. During the practical sessions students will learn how to reduce the amount of code you write to build rich user interface applications, Modularizing code, retrieving data from back-end- server and manipulate it.

**Outline of the course:**

| Unit Number | Title of the Unit                                     | Minimum Number of Hours |           |
|-------------|-------------------------------------------------------|-------------------------|-----------|
|             |                                                       | Theory                  | Practical |
| 1           | Introduction to JavaScript and JavaScript Frameworks. | 04                      | 36        |
| 2           | Directives in AngularJS                               | 07                      |           |
| 3           | MVW-The AngularJS way                                 | 06                      |           |
| 4           | Introduction to NodeJS                                | 07                      |           |
| 5           | Working with ExpressJS Framework                      | 06                      |           |
| 6           | Working with Database                                 | 06                      |           |

**Total Hours (Theory): 36**

**Total Hours (Lab) : 36**

**Total Hours: 72**

## **Detail Syllabus:**

### **Unit I: Introduction to JavaScript and JavaScript Frameworks      Hours 04**

Introduction JavaScript and its DOM concepts, Arrow function, JSON Objects, JavaScript Frameworks & Libraries, MEAN.JS introduction, Architecture of MEAN.JS.

### **Unit II: Introduction to AngularJS      Hours 07**

Introduction to AngularJS, Features of AngularJS, Data Binding in angularJS, working with Expressions in angularJS, Introduction to Directives, Directive Lifecycle, Using Angular JS built-in directives, creating a custom directive, Overview of \$scope, \$rootScope.

### **Unit III : MVW-The Angular JS way      Hours 06**

MVW Architecture: Model-View-Controller and Model-View-View-Model Architecture, Introduction to AngularJS Modules – Application module & Controller modules, Attaching Properties and functions to scope, Controller in external files, AngularJS Filters, working with Angular Forms, Form events, validating Angular forms.

### **Unit IV : Introduction to NodeJS      Hours 07**

Introduction to NodeJS, Features of NodeJS, Traditional web Server Model, Node.js process Model, Installation of NodeJS, working with Node Package Manager, Command Line options, NodeJS modules – Core Modules, Creating Local modules & Exporting Local modules, Third-party modules, Creating NodeJS web server.

### **Unit V : Introduction to ExpressJS Framework      Hours 06**

Introduction to ExpressJS, Installing ExpressJS using NPM, Advantages of ExpressJS, Installing Express.js, building web server, configure routes, ExpressJS middleware's, Serve Static Resources using Express.js, working with HTTP methods of ExpressJS, ExpressJS routing, Creating RESTful API.

### **Unit IV : Working with Database      Hour 07**

Introduction to Mongo DB, Installation of MongoDB – MongoDB Server & MongoDB Shell, MongoDB Compass, MongoDB Database, Collections, and documents, Basic operations using MongoDB – Create, Insert, Update and Delete, Access MongoDB in Node.js, setting up mongoose, Connecting MongoDB, Insert, update and delete document.

**Core Books:**

1. Jeffry Houser : “Learn With: Angular 5, Bootstrap, and NodeJS”, Kindle Edition,
2. Shyam Seshadri Brad Green: “AngularJS – Up and Running, Brad Green”, Second Edition, O'REILLY
3. Basarat Ali Syed: “Beginning Node.js”, Apress Publication.
4. Greg Lim : “Beginning MEAN Stack (MongoDB, Express, Angular, Node.js)”, kindle Edition.

**Reference Books:**

1. Agus Kurniawan: “AngularJS Programming by Example 2017 Edition”, Kindle Edition.
2. Adam Freeman: “ Pro AngularJS 2017 Edition”, Apress.
3. Krasimir Tsonev: “Node.js By Example”, Packt Publishing
4. Ethan Brown: “Web Development with Node and Express”, O'REILLY

**Web References:**

1. <http://www.w3schools.com/angular/default.asp> [Tutorial link for AngularJS]
2. <http://www.tutorialspoint.com/angularjs/> [Tutorial link for AngularJS]
3. [https://www.tutorialspoint.com/angularjs/angularjs\\_tutorial.pdf](https://www.tutorialspoint.com/angularjs/angularjs_tutorial.pdf) [E-book for AngularJS]
4. <http://www.tutorialsteacher.com/nodejs/nodejs-modules> [Tutorial link for NodeJS]
5. <https://www.javatpoint.com/mean-stack-tutorial> [MEAN stack development]

**Course Outcomes:** Upon successful completion of the course, students will be,

|       |                                                                                                     |
|-------|-----------------------------------------------------------------------------------------------------|
| CO1 : | Familiar with JavaScript and JavaScript frameworks.                                                 |
| CO2 : | Able to understand basic fundamental of AngularJS and implement Model-View-View-Model Architecture. |
| CO3 : | Able to create NodeJS and ExpressJS applications.                                                   |
| CO4 : | Able to create Restful API to access data from MongoDB.                                             |
| CO5 : | Able to create Single page application and MEAN stack application.                                  |

### Course Outcomes Mapping:

| Unit No. | Unit Name                                            | Course Outcomes |     |     |     |     |
|----------|------------------------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                                      | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | Introduction to JavaScript and JavaScript Frameworks | √               |     |     |     | √   |
| 2        | Introduction to AngularJS                            |                 | √   |     |     | √   |
| 3        | MVW-The Angular JS way                               |                 | √   |     |     | √   |
| 4        | Introduction to NodeJS                               |                 |     | √   | √   | √   |
| 5        | Introduction to ExpressJS Framework                  |                 |     | √   | √   | √   |
| 6        | Working with Database                                |                 |     |     | √   | √   |

### Course Articulation Matrix:

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | -   | -   | -   | -   | 2   | -   | 1   | -   | -   | -    | -    | 1    | -    | -    |
| CO2 | 1   | 2   | 2   | 1   | 2   | -   | 1   | 1   | -   | 1    | 1    | 2    | 1    | 2    |
| CO3 | 1   | -   | 3   | 1   | 3   | -   | 2   | 1   | -   | -    | 1    | 2    | 1    | 1    |
| CO4 | 2   | 2   | 3   | 2   | 3   | 2   | 2   | 3   | -   | 2    | 2    | 3    | 3    | 3    |
| CO5 | 2   | 3   | 3   | 2   | 3   | 2   | 2   | 3   | 2   | 2    | 3    | 3    | 3    | 2    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation

## HSUA501 - ACADEMIC WRITING

### Credits and Hours:

| Teaching Scheme | Theory | Practical | Tutorial | Total | Credit |
|-----------------|--------|-----------|----------|-------|--------|
| Hours/week      | --     | 30/15     | --       | 30/15 | 2      |
| Marks           | --     | 100       | --       | 100   |        |

### Pre-requisite courses:

- An Intermediate Guide to Writing in English for University Study  
<https://www.futurelearn.com/courses/english-for-study-intermediate/4/todo/62943>

### Outline of the Course:

| Sr. No. | Title of the unit                          | Minimum number of hours |
|---------|--------------------------------------------|-------------------------|
| 1.      | Academic Writing and Research Process      | 05                      |
| 2.      | Anatomy of Academic Writing                | 05                      |
| 3.      | Key Academic Skills                        | 05                      |
| 4.      | Accuracy in Academic Writing               | 05                      |
| 5.      | Using and Citing Sources of Ideas          | 05                      |
| 6.      | Contemporary Practices in Academic Writing | 05                      |

**Total hours (Practical): 30**

**Total hours :30**

**Detail Syllabus:**

- |                                                                                                                                                                                                     |         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1. Academic Writing and Research Process                                                                                                                                                            | 5 Hours |
| Introduction to Academic Writing, Academic Writing as a Part of Research, Types of Academic Writing, Features of Academic Writing, Importance of Good Academic Writing in various Academic Works    |         |
| 2. Anatomy of Academic Writing                                                                                                                                                                      | 5 Hours |
| Academic Vocabulary, Simple and Complex Sentences, Organizing Paragraphs, The Writing Process, Adopting Academic Writing Style                                                                      |         |
| 3. Key Academic Skills                                                                                                                                                                              | 5 Hours |
| Note – taking, Note – making, Paraphrasing, Summarizing                                                                                                                                             |         |
| 4. Accuracy in Academic Writing                                                                                                                                                                     | 5 Hours |
| Lexical Range, Academic Language and Structures, Elements of Writing, Proof Reading, Editing, and Rewriting                                                                                         |         |
| 5. Using and Citing Sources of Ideas                                                                                                                                                                | 5 Hours |
| Academic Texts and their Types, Intellectual Honesty in Academic Writing, Avoiding Plagiarism – Idea Theft, Degrees of Plagiarism, Types of Borrowing, Anatomy of Citations, Common Citation Styles |         |
| 6. Contemporary Practices in Academic Writing                                                                                                                                                       | 5 Hours |
| Analytical Essays, Graph / Table / Process Interpretation and Description, Writing Reports and Abstract, Writing Research / Concept Papers                                                          |         |

### Course Outcome (COs):

At the end of the course, the students

|     |                                                                                      |
|-----|--------------------------------------------------------------------------------------|
| CO1 | Will have sound understanding of the concept and applications of academic writing    |
| CO2 | Will have acquired enough knowledge of academic writing style, strategy and approach |
| CO3 | Will be able to demonstrate error free and effective academic writing                |
| CO4 | Will be able to demonstrate ability to work on project/report/paper writing          |
| CO5 | Will have the sound understanding of the Research and Research Methodology           |
| CO6 | Will be effectively communicating in diverse academic and professional settings.     |

### Course Articulation Matrix:

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | -   | -   | -   | -   | -   | -   | -   | -   | 3   | -    | -    | -    | -    | -    |
| CO2 | -   | -   | -   | -   | -   | -   | -   | -   | 3   | -    | -    | -    | -    | -    |
| CO3 | -   | -   | -   | -   | -   | -   | -   | -   | 3   | -    | -    | -    | -    | -    |
| CO4 | -   | -   | -   | -   | -   | -   | -   | -   | 3   | -    | -    | -    | -    | -    |
| CO5 | -   | -   | -   | -   | -   | -   | 3   | -   | 3   | -    | -    | -    | -    | 2    |
| CO6 | -   | -   | -   | -   | -   | -   | 2   | -   | -   | -    | -    | -    | -    | -    |

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

**Recommended Study Material:**

**Text book:**

1. Academic Writing for International Students, Routledge
2. Academic Writing: A Guide for Management Students and Researchers. Monipally, M.M. & Pawar, B.S. Sage. 2010. New Delhi
3. Effective Academic Writing Level - 1,2,3,4 (Second Edition) By: Alice Savage, Patricia Mayer, Masoud Shafiei, Rhonda Liss, & Jason Davis; Publisher: Oxford

**Reference book:**

1. Writing Your Thesis (2nd Edition) by Paul Oliver, Sage
2. Development Communication In Practice by Vilanilam V J, Sage
3. Intercultural Communication by Mingsheng Li, Patel Fay, Sage

**Web material:**

1. [www.owl.perdue.edu](http://www.owl.perdue.edu)

# **TEACHING SCHEME & DETAILED SYLLABUS**

**FOR**

**M.Sc. (IT) PROGRAMME  
(3<sup>rd</sup> SEMESTER)**

**EFFECTIVE FROM  
ACADEMIC YEAR 2025-26**

**Teaching & Examination Scheme**  
**Master of Science in Information Technology (M. Sc. (IT)) Programme**  
**(Choice Based Credit System)**

**Effective from Year 2025 - 26**  
**Semester - III**

| Course Code     | Course Title                           | Teaching Scheme |           |       |        | Examination Scheme |          |           |          |       |     |
|-----------------|----------------------------------------|-----------------|-----------|-------|--------|--------------------|----------|-----------|----------|-------|-----|
|                 |                                        | Contact Hours   |           |       | Credit | Theory             |          | Practical |          | Total |     |
|                 |                                        | Theory          | Practical | Total |        | Internal           | External | Term Work | External |       |     |
| CTUE601-CTUE603 | Elective-III                           | 3               | 3         | 6     | 6      | 50                 |          | 50        | 50       | 50    | 200 |
| CTUC604         | Software Quality Assurance and Testing | 3               | 3         | 6     | 6      | 50                 |          | 50        | 50       | 50    | 200 |
| CTUC605         | Mini Project                           | -               | 12        | 12    | 12     | -                  | -        | -         | 200      | 200   | 400 |
| CLUV601         | Environmental Sciences                 | -               | 2         | 2     | 2      | -                  | -        | -         | 50       | 50    | 100 |
|                 |                                        | 06              | 20        | 26    | 26     | 200                |          | 700       |          | 900   |     |

| Elective-III |                           |
|--------------|---------------------------|
| Course Code  | Course Title              |
| CTUE601      | Advanced Game Development |
| CTUE602      | Data Analytics            |
| CTUE603      | Internet of Things        |

## CTUE601 - Advanced Game Development (200 Marks)

**Contact Hours: 06**

**Pre-requisite:** Basics of Animation, C# Language, Mathematics and Physics concepts

**Methodology & Pedagogy:** During theory lectures illustrations Graphics, animation and various concepts regarding Game Development. Emphasize will be given on some mathematical and physics concepts, Fundamental of Graphics and objects creation, 3D graphics, Collision detections, Fundamental of Game programming, Game loop, Game Engine and many more. During Practical sessions, students will develop 3D games using Unity.

**Outline of the course:**

| Unit Number | Title of the Unit                          | Minimum Number of Hours |           |
|-------------|--------------------------------------------|-------------------------|-----------|
|             |                                            | Theory                  | Practical |
| 1           | Introduction to Unity for Game Development | 06                      | 36        |
| 2           | Tools and Resources for Game Development   | 06                      |           |
| 3           | Scene Designing in Gaming                  | 06                      |           |
| 4           | Interface Designing in Gaming              | 06                      |           |
| 5           | Rendering and Walkthrough                  | 06                      |           |
| 6           | Testing and Deployment of Game             | 06                      |           |

**Total Hours (Theory): 36**

**Total Hours (Lab): 36**

**Total Hours: 72**

## **Detail Syllabus:**

### **Unit I: Introduction to Unity for Game Development**

**Hours 06**

Getting started with 3D, Coordinates, Local space versus World space, Vectors, Cameras, Polygons, edges, vertices, and meshes, Materials, textures, Shaders, Rigid Body physics, Collision detection. Essential Unity concepts, Introduction to Assets, Scenes, Game Objects, Components, Scripts, Prefabs. The Interface: Scene window and Hierarchy, Inspector, Project and Game window.

### **Unit II: Tools and Resources for Game Development**

**Hours 06**

The terrain editor: menu features, importing and exporting height maps, Terrain Toolset: Raise, plain and smooth the height, Paint texture, place trees, and terrain settings, creating island: use of Sun, Sea and Sand. Importing Model package. Player characters: working with inspector, Tags, Layers, Prefabs and the Inspector, Deconstructing the First Person Controller object, Parent-child issues, First Person Controller objects. Scripting basics: Variables, Functions, If Else, comments and major syntax. The FPS walker script

### **Unit III: Scene Designing in Gaming**

**Hours 06**

Discovering Collisions, ray casting, Adding the outpost, Opening the outpost, Collision detection and creating new assets, attaching a script. Ray casting with disabling collision detection. Prefab. Collection and HUD: Creating the battery prefab. Download, import, and place, Tagging the battery, Scale, collider, and rotation, scattering batteries, Displaying the battery GUI, Creating the GUI Texture object, Positioning the GUI Texture, Scripting for GUI change, Battery collection with triggers, Restricting outpost access, Restricting access Hints for the player, Creating the fire particle systems

### **Unit IV: Interface Designing in Gaming**

**Hours 06**

Interfaces and menus, Making the main menu, Adding the play button, Disabling Game Objects, Writing an OnGUI() script for a simple menu, Flexible positioning for GUIs, Adding UnityGUI buttons, Opening scenes with custom functions, GUI skin settings. Decision time

### **Unit V: Rendering and Walkthrough**

**Hours 06**

Downloading assets, Making the smoke material, Particle system settings, Ellipsoid Particle Emitter settings, Particle Animator settings, Adding audio to the volcano, Volcano testing, Coconut trails, Editing the Prefab, Trail Renderer component, Updating the prefab, Performance tweaks, Camera Clip Planes and fog, Ambient lighting Instructions scene, Adding screen text, Text Animation using Linear Interpolation (Lerp), Menu return, Island level fade-in, UnityGUI texture rendering, Game win notification.

## **Unit VI: Testing and Deployment of Game**

**Hours 06**

Build Settings, Web Player, Player Settings, Web Player Streamed, OS X Dashboard Widget, OS X/Windows, Standalone, Building the game, adapting for web build, Texture compression and debug stripping, building standalone 266Indie versus Pro, building for the Web, adapting web player builds, Quality Settings, Player Input settings, sharing your work, Testing and finalizing: Public testing

### **Core Books:**

1. Will Goldstone: Unity Game Development Essentials: Edition 2<sup>nd</sup>: Packt Publication: 2009.
2. Greg Lukosek, "Learning C# by Developing Games with Unity 5.x", 2nd Edition, Packt Publishing, 2016
3. Ashley Godbold, Simon Jackson, "Mastering Unity 2D Game Development", 2nd Edition, Packt Publishing, 2016.
4. Joe Hocking, "Unity in Action: Multiplatform game development in C#", 2nd Edition, Manning Publications, 2018
5. Sue Blackman, "Beginning 3D Game Development with Unity 4: All-in-one, multi-platform game development", 2nd Edition, Apress, 2013

### **Reference Books:**

1. Nicolas Alejandro Borrromeo: Hands-On Unity 2021, Game Development: Edition 2<sup>nd</sup>: Packt Publication: 2021.
2. Ben Tristem, Mike Geig, "Unity Game Development in 24 Hours- Sams Teach Yourself", 2nd Edition, Paperback, December 19, 2015
3. Michelle Menard, "Game Development with Unity", Course Technology, 2013.
4. Matt Smith, Chico Queiroz, "Unity 5.x Cookbook", Packt Publishing, October 5, 2015
5. Francesco Sapio, "Unity UI Cookbook", Packt Publishing, 2015
6. P Patrick Felicia, "Unity 5 from Zero to Proficiency: A step-by-step guide to creating your first game", CreateSpace Independent Publishing Platform, February 25, 2016
7. Matt Smith, "Unity 2018 Cookbook", 3rd Edition, Packt Publishing, August 31, 2018
8. Alan Thorn, "Mastering Unity Scripting", Packt Publishing, January 29, 2015

### Web References:

1. <https://subscription.packtpub.com/search?query=unity>
2. <https://itsourcecode.com/free-projects/python-projects/mario-game-in-python-with-source-code/>
3. <http://learn.unity.com>
4. <https://unity3d.com/learn/tutorials/topics/developer-advice/how-start-your-game-development> [Game Tutorial]
5. <https://www.studytonight.com/game-development-in-2D/> [Game Tutorial]
6. <https://msdn.microsoft.com/en-us/magazine/dn759441.aspx> [Game Tutorial]

**Course Outcomes:** Upon successful completion of the course, students will be,

|       |                                                                               |
|-------|-------------------------------------------------------------------------------|
| CO1 : | able to gain a basic understanding of game development using Unity 3D         |
| CO2 : | able to learn concepts such as manipulating objects, scripting, and compiling |
| CO3 : | able to learn graphics and visuals in game development                        |
| CO4 : | able to develop script writing for any problem and solution as game           |
| CO5 : | able to develop full 3D game                                                  |

### Course Outcomes Mapping:

| Unit No. | Unit Name                                  | Course Outcomes |     |     |     |     |
|----------|--------------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                            | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | Introduction to Unity for Game Development | √               |     |     |     |     |
| 2        | Tools and Resources for Game Development   |                 | √   |     |     |     |
| 3        | Scene Designing in Gaming                  |                 | √   |     |     |     |
| 4        | Interface Designing in Gaming              |                 |     | √   |     |     |
| 5        | Rendering and Walkthrough                  |                 |     |     | √   |     |
| 6        | Testing and Deployment of Game             |                 |     |     |     | √   |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 3   | 2   | -   | 1   | 2   | -   | 2   | 1   | 1   | -    | 1    | 1    | 2    | 3    |
| CO2 | 3   | 2   | -   | 2   | 2   | -   | 2   | 1   | 1   | -    | 1    | 1    | 2    | 1    |
| CO3 | 3   | 1   | 1   | 3   | 2   | -   | 2   | 1   | 1   | -    | 2    | 2    | 2    | 1    |
| CO4 | 3   | 2   | 2   | 2   | 2   | 1   | 2   | 3   | 1   | 2    | 3    | 3    | 3    | 2    |
| CO5 | 3   | 3   | 3   | 2   | 2   | 1   | 2   | 3   | 1   | 3    | 3    | 3    | 3    | 3    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## CTUE602 - Data Analytics (200 Marks)

**Contact Hours: 06**

**Pre-requisite:** CA845: Advanced Database Technologies

**Methodology & Pedagogy:** During theory lectures the emphasis will be given on the basics of data analytics and related tools and techniques. Students will be introduced to the concepts of data science, exploratory data analysis, supervised and unsupervised learning methods. Applications as well as research trends and future direction of data analytics will be discussed with the length. During the practical sessions, students will be introduced to tools of data analytics such as R and Python. Students will be given appropriate case studies of data analytics to get the real time exposure of data analytics.

**Outline of the course:**

| Unit Number | Title of the Unit                          | Minimum Number of Theory Hours | Practical |
|-------------|--------------------------------------------|--------------------------------|-----------|
| 1           | Introduction of Data Analytics             | 06                             | 36        |
| 2           | Exploratory Data Analysis                  | 06                             |           |
| 3           | Supervised & Unsupervised Learning Methods | 06                             |           |
| 4           | Deep Re-enforcement Learning               | 06                             |           |
| 5           | Data visualization                         | 06                             |           |
| 6           | Recent trends and future directions        | 06                             |           |

**Total Hours (Theory): 36**

**Total Hours (Lab): 36**

**Total Hours: 72**

## **Detail Syllabus:**

### **Unit I : Introduction of Data Analytics** **Hours 06**

What is Data Science? ,Big Data and Data Science hype ,Why now? , Datafication , Current landscape of perspectives , Skill sets needed, Data Analysis cycle, Data analysis Vs Data analytics, Life cycle of data

### **Unit II : Exploratory Data Analysis** **Hours 06**

Philosophy of EDA - The Data Science Process ,Statistical Inference , Populations and samples , Statistical modeling, probability distributions, fitting a model Introduction to R ,Basic tools (plots, graphs and summary statistics) of EDA, SQL with data life cycle, Python for EDA.

### **Unit III : Supervised & Unsupervised Learning Methods** **Hours 06**

Supervised methods: Linear Regression , Classification Trees, Random Forest, Clustering, Support Vector Machine, Regularization, Radial Basis, Association Rule Mining, Dimensionality Reduction - Singular Value Decomposition – Principal Component Analysis

### **Unit IV : Deep Re-enforcement Learning** **Hours 06**

Introduction to Neurons, Activation functions, single and multi-layer perceptron, back propagation, Application, Hyper-parameters and Estimation, Gradient Descent, Curiosity driven learning.

### **Unit V : Data visualization** **Hours 06**

Introduction of data visualization, Power BI, and Tableau, ggplot+ggplot2, Seaborn.

### **Unit VI : Recent trends and future directions** **Hours 06**

Data Visualization - Basic principles, ideas and tools for data visualization, Data Science and Ethical Issues - Discussions on privacy, security, ethics, Next-generation data scientists, Basics of Big Data analytics

## **Core Books:**

1. Cathy O'Neil and Rachel Schutt: Doing Data Science, Straight Talk From The Frontline, O'Reilly. 2014.
2. Jure Leskovec, Anand Rajaraman, and Jeffrey David Ullman, Mining of Massive Datasets Cambridge University Press,2nd Edition, New York, NY, USA,2014.
3. Howard B. Demuth, Mark H. Beale, Orlando De Jess, and Martin T. Hagan, Neural Network Design , paperback USA, 2nd Edition,2014.

**Reference Books:**

1. Walpole, R. E., Myers, R. H., Myers, S. L., & Ye, K. , Probability & statistics for engineers & scientists ,9th edition, Prentice Hall,2012.
2. Haykin, S. S., Haykin, S. S., Haykin, S. S., & Haykin, S. S., Neural networks and learning machines Pearson, Vlolume 3,2009.
3. Mohammed J. Zaki and Wagner Miera Jr, Data Mining and Analysis: Fundamental Concepts and Algorithms.,Cambridge University Press. 2014.

**Web References:**

1. [https://onlinecourses.nptel.ac.in/noc17\\_mg24/preview](https://onlinecourses.nptel.ac.in/noc17_mg24/preview) [ Online Data Analytics Course ]
2. <https://www.itl.nist.gov/div898/handbook/eda/section1/eda11.htm> [ Exploratory Data AnalysisMaterial]
3. [https://datahoarder.io/Humble%20Bundle%20Books/Humble%20Book%20Bundle\\_%20Data%20Science%20presented%20by%20O\\_Reilly/doingdatascience.pdf](https://datahoarder.io/Humble%20Bundle%20Books/Humble%20Book%20Bundle_%20Data%20Science%20presented%20by%20O_Reilly/doingdatascience.pdf) [ Data Science E-Book]
4. [http://www.astro.caltech.edu/~george/aybi199/Donalek\\_Classif.pdf](http://www.astro.caltech.edu/~george/aybi199/Donalek_Classif.pdf) [ Supervised andunsupervised methods tutorials]
5. <https://yourstory.com/2017/12/data-analytics-future-trends/> [ Future Trends of Data Analytics]
6. [http://www.astro.caltech.edu/~george/aybi199/Donalek\\_Classif.pdf](http://www.astro.caltech.edu/~george/aybi199/Donalek_Classif.pdf) [ Supervised andunsupervised methods tutorials]
7. <https://yourstory.com/2017/12/data-analytics-future-trends/> [ Future Trends of Data Analytics]

**Course Outcomes:** Upon successful completion of the course, students will be,

|       |                                                                                                     |
|-------|-----------------------------------------------------------------------------------------------------|
| CO1 : | Describe what Data Science and Data Analytics are and the skill sets needed to be a data scientist. |
| CO2 : | Understand significance of exploratory data analysis in statistical and visualization aspects.      |
| CO3 : | Understand and apply data analytics techniques such as supervised, unsupervised and EDA.            |
| CO4 : | Understand the importance of self-learning environments                                             |
| CO5 : | Able to understand the recent trends and future directions of data analytics.                       |

**Course Outcomes Mapping:**

| Unit No. | Unit Name                                  | Course Outcomes |     |     |     |     |
|----------|--------------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                            | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | Introduction of Data Analytics             | √               |     |     |     |     |
| 2        | Exploratory Data Analysis                  |                 | √   |     |     |     |
| 3        | Supervised & Unsupervised Learning Methods |                 |     | √   |     |     |
| 4        | Deep Re-enforcement Learning               |                 |     |     | √   |     |
| 5        | Data visualization                         |                 |     |     | √   |     |
| 6        | Recent trends and future directions        |                 |     |     |     | √   |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 3   | 2   | 2   | 1   | -   | -   | -   | 1   | -   | -    | -    | 2    | 3    | 1    |
| CO2 | 2   | 2   | 1   | 1   | -   | 1   | -   | -   | -   | -    | -    | 2    | 2    | 1    |
| CO3 | 1   | 3   | 3   | 2   | -   | 1   | 3   | -   | -   | -    | -    | 2    | 1    | 1    |
| CO4 | 2   | 1   | 2   | 2   | 2   | 1   | 2   | -   | -   | -    | -    | 2    | 1    | 2    |
| CO5 | 1   | 1   | 2   | 3   | 1   | -   | 2   | -   | -   | -    | -    | 3    | 2    | 1    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## CTUE603 - Internet of Things (200 Marks)

**Contact Hours: 06**

**Pre-requisite: -**

**Methodology & Pedagogy:** In order to achieve the objectives and goals, students will be taught the basics of IOT with its structure, layers and applications. Fundamentals controllers are introduced which can be used to implement the IOT based projects. Students will also be introduced to interface GPIO with controllers like arduino and Raspberry PI. They can also learn to develop a desktop application and mobile application which can control the device remotely.

### Outline of the course:

| Unit Number | Title of the Unit                                   | Minimum Number of Hours |           |
|-------------|-----------------------------------------------------|-------------------------|-----------|
|             |                                                     | Theory                  | Practical |
| 1           | Introduction to Internet of Things                  | 06                      | 36        |
| 2           | Programming Internet of Things System               | 08                      |           |
| 3           | Sensors and Actuators for Internet of Things System | 04                      |           |
| 4           | IoT Connectivity Techniques                         | 06                      |           |
| 5           | ESP8266 and ESP32 Microcontroller for IOT           | 06                      |           |
| 6           | Raspberry Pi for IOT Projects                       | 06                      |           |

**Total Hours (Theory): 36**

**Total Hours (Lab) :36**

**Total Hours:72**

## **Detail Syllabus:**

### **Unit I : Introduction to Internet of Things**

**Hours: 06**

Introduction - Overview of Internet of Things (IoT), The building blocks of IoT, Various technologies making up IoT ecosystem, IoT levels, IoT design methodology, The Physical Design and Logical Design of IoT, Functional blocks of IoT and Communication Models, Development Tools used in IoT.

### **Unit II : Programming Internet of Things System**

**Hours: 08**

Arduino / node MCU controller for IOT Programming, Embedded C data types, variables, constants and operators, programming using control statements, loops, string and array, Arduino IDE. Variables and Numbers, Looping Structures, Conditional Statements, Lists, Tuples and Dictionaries, Type Conversions, Function declaration, calling functions and passing values, Function Returning values. Exiting from functions.

### **Unit III : Sensors and Actuators for Internet of Things System**

**Hours: 04**

Interfacing with various sensors like temperature sensor, PIR sensor and ultrasonic sensor. Interfacing with servomotor, DC motor, Gas sensor.

### **Unit IV : IoT Connectivity Techniques**

**Hours: 06**

RFID, RFID Features, IoT Protocols based on Functionalities, 6LoWPAN, WiFi, Bluetooth, LPWAN, Data Protocols, MQTT, MQTT Components, MQTT Methods, MQTT Applications, SMQTT, CoAP, CoAP Message Types, CoAP Request Response Model.

### **Unit V : ESP8266 and ESP32 Microcontroller for IOT**

**Hours: 06**

Basics of Wireless Networking, Introduction to ESP8266 Wi-Fi Module, Various Wi-Fi library, Web server- introduction, installation, configuration, Posting sensor(s) data to web server, Introduction to ESP32 microcontroller.

### **Unit VI : Raspberry Pi for IOT Projects**

**Hours: 06**

Introduction to Raspberry Pi, Why Raspberry pi, Features of Raspberry pi, Basic set up and first boot configuration, Different uses of Raspberry pi, Different Versions of Raspberry pi, Vision of Raspberry pi, Different components of the Board.

## **Core Books:**

1. Arshdeep Bahga, Vijay Madisetti: Internet of Things: A Hands-On Approach, VPT Publication, 2014.
2. Dr. Simon Monk: Programming the Raspberry Pi: Getting Started with Python, McGraw Hill Publication

### Reference Books:

1. M. Richardson, S. Wallace: Getting Started with Raspberry Pi, O'Reilly, 2012
2. Dr. Simon Monk: Programming the Raspberry Pi: Getting Started with Python, McGraw Hill Publication

### Web References:

1. <https://www.tutorialspoint.com/arduino/index.htm> [Arduino Fundamentals]
2. [www.tinkercad.com](http://www.tinkercad.com) [Arduino Simulator]
3. <https://www.youtube.com/watch?v=UUOChoCbty8> [Raspberry Pi - How to start programming with Python]
4. <https://www.raspberrypi.org/documentation/usage/python/> [Step by Step Programming]

**Course Outcomes:** Upon successful completion of the course, students will be,

|       |                                                                 |
|-------|-----------------------------------------------------------------|
| CO1 : | Ability to understand IOT Terminology                           |
| CO2 : | Understanding IOT Programming                                   |
| CO3 : | Understanding use of sensors and actuators                      |
| CO4 : | Learning IoT Connectivity methods                               |
| CO5 : | Designing IOT projects using ESP8266 and ESP32 and Raspberry PI |

### Course Outcomes Mapping:

| Unit No. | Unit Name                                           | Course Outcomes |     |     |     |     |
|----------|-----------------------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                                     | CO1             | CO2 | CO3 | CO4 | CO5 |
| 1        | Introduction to Internet of Things                  | √               |     |     |     |     |
| 2        | Programming Internet of Things System               |                 | √   |     |     |     |
| 3        | Sensors and Actuators for Internet of Things System |                 |     | √   |     |     |
| 4        | IoT Connectivity Techniques                         |                 |     |     | √   |     |
| 5        | ESP8266 and ESP32 Microcontroller for IOT           |                 |     |     |     | √   |
| 6        | Raspberry Pi for IOT Projects                       |                 |     |     |     | √   |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 3   | 1   | 1   | 3   | 1   | 1   | 1   | 1   | 1   | -    | -    | 1    | -    | -    |
| CO2 | 3   | 1   | 2   | 3   | 1   | 1   | 1   | 2   | 2   | -    | -    | 1    | 1    | -    |
| CO3 | 2   | 2   | 2   | 1   | 1   | 1   | 2   | 2   | 2   | -    | -    | 2    | 1    | 1    |
| CO4 | 2   | 2   | 3   | 2   | 1   | 2   | 2   | 2   | 2   | -    | -    | 2    | 1    | 1    |
| CO5 | 1   | 2   | 3   | 2   | 1   | 2   | 3   | 2   | 2   | -    | -    | 2    | 1    | 3    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## **CTUC604 - Software Quality Assurance and Testing**

### **(200 Marks)**

**Contact Hours: 06**

**Pre-requisite:** Software Engineering.

**Methodology & Pedagogy:** The theory sessions will cover the software testing concepts and practices that support the production of quality software. The practical sessions will cover the application of testing techniques at various levels of testing using manual and automated testing tools. This is

### **Outline of the Course:**

| <b>Unit No.</b> | <b>Title of the Unit</b>           | <b>Minimum Numbers of Hours</b> |                  |
|-----------------|------------------------------------|---------------------------------|------------------|
|                 |                                    | <b>Theory</b>                   | <b>Practical</b> |
| 1               | Basic concepts of Software Testing | 05                              | 36               |
| 2               | White Box Testing Techniques       | 05                              |                  |
| 3               | Black Box Testing Techniques       | 07                              |                  |
| 4               | Levels of Testing                  | 07                              |                  |
| 5               | Test Planning and Design           | 06                              |                  |
| 6               | Test Execution and Measurements    | 06                              |                  |

**Total Hours (Theory): 36**

**Total Hours(Lab): 36**

**Total hours : 72**

### **Detail Syllabus:**

**Unit – I: Basic concepts of Software Testing**

**Hours: 05**

Introduction to testing and its importance in software development, software testing role, responsibilities and principles, concept of quality, Verification and Validation, Design test case, Introduction to Static Techniques and Dynamic test case design techniques.

**Unit – II: White Box Testing Techniques**

**Hours: 05**

Overview of White Box, Control flow testing: Statement Coverage Testing, Branch Coverage Testing, Path Coverage Testing, Conditional Coverage Testing  
Data flow testing: Data Flow Anomaly, Overview of Dynamic Data Flow Testing, Data Flow Graph, Data Flow Terms, Data Flow Testing Criteria.

**Unit – III: Black Box Testing Techniques**

**Hours: 07**

Overview of Black Box: Equivalence Class Partition, Boundary Value Analysis, Pairwise Technique, Cause Effective Graph, Decision Table.

**Unit – IV: Levels of Testing**

**Hours: 07**

Introduction to functional and non functional testing  
Functional Testing : Unit Testing, Integration Testing, System Testing, User Acceptance Testing. Sanity/Smoke Testing, Regression Test.  
Non Functional Testing: Performance Testing, Scalability Testing, Interoperability Testing, Security Testing, Recovery Testing, Basic Testing.

**Unit – V: Test Planning and Design**

**Hours: 06**

Requirements Analysis/Design : Understand the requirements, Prepare Traceability Matrix Test Planning: Object, Scope of Testing, Schedule, Approach, Roles & Responsibilities, Assumptions, Risks & Mitigations, Entry & Exit Criteria, Test Automation, Deliverables. Test Cases Design: Write Test cases, Review Test cases, Test Cases Template, Types of Test Cases, Difference between Test Scenarios and Test Cases. Test Environment setup: Understand the SRS, Hardware and software requirements, Test Data

**Unit – VI: Test Execution and Measurements**

**Hours: 06**

Test Execution: Execute test cases, Defect Tracking and Reporting: Types of Bugs, Identifying the Bugs, Bug/Defect Life Cycle, Reporting the Bugs, Severity and priority.  
Test Closure: Criteria for test closure, Test summary report  
Test Metrics: Test Measurements, significance of Test Metrics, Metric Life Cycle, Types of Manual Test Metrics.

**Core Books:**

1. Sagar Naik, Piyu Tripathy: Software Testing and Quality Assurance, Theory and Practice, Wiley, 2008.
2. Paul C. Jorgensen : Software Testing: A Craftsman's Approach, 4th Edition by , CRC press, Taylor and Francis Group, 2014
3. Roger S Pressman: Software Engineering – A Practitioner's Approach, 7th Edition, McGRAW HILL International Edition, 2010.

### Reference Books:

1. Mauro Pezze, Michael Young: Software testing and Analysis- Process, Principles and Techniques, Wiley India, 2012.
2. Boris Beizer: Software Testing Techniques: 2nd Edition, Van Nostrand Reinhold, 1990.
3. Daniel Galin: Software Quality Assurance, Pearson Education, 2004.
4. Ron Patton: Software Testing, Pearson Education, 2005.

### Web References:

1. <http://tryqa.com/what-is-software-testing> [Basic of Software Testing ]
2. <https://www.guru99.com/functional-testing.html> [Functional Testing]
3. <http://www.softwaretestinggenius.com/download/bgstpadmini.pdf> [ Software Testing Life Cycle]

**Course Outcomes:** Upon successful completion of the course, the students will:

|       |                                                         |
|-------|---------------------------------------------------------|
| CO1 : | Understand the role of testing in software development. |
| CO2 : | Apply the test case design techniques.                  |
| CO3 : | Acquire the various levels of testing.                  |
| CO4 : | Document test plan and design.                          |
| CO5 : | Execute test case and measure the results.              |

**Course Outcomes Mapping :**

| Unit No. | Unit Name                          | Course Outcomes |     |     |     |     |
|----------|------------------------------------|-----------------|-----|-----|-----|-----|
|          |                                    | Co1             | Co2 | Co3 | Co4 | Co5 |
| 1        | Basic concepts of Software Testing | √               |     |     |     |     |
| 2        | White Box Testing Techniques       | √               | √   |     |     |     |
| 3        | Black Box Testing Techniques       | √               | √   |     |     |     |
| 4        | Levels of Testing                  | √               |     | √   |     |     |
| 5        | Test Planning and Design           | √               |     |     | √   |     |
| 6        | Test Execution and Measurements    | √               |     |     |     | √   |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 3   | 2   | 3   | 2   | 2   | 3   | 2   | 2   | 3   | 2    | 2    | 2    | 3    | 2    |
| CO2 | 3   | 2   | 2   | 2   | 3   | 2   | 2   | 1   | 1   | 2    | 1    | 1    | 1    | 2    |
| CO3 | 3   | 2   | 2   | 1   | 2   | 3   | 2   | 1   | 1   | 1    | 2    | 1    | 1    | 2    |
| CO4 | 3   | 2   | 1   | 1   | 2   | 2   | 2   | 2   | 1   | 2    | 1    | 2    | 1    | 1    |
| CO5 | 3   | 3   | 2   | 1   | 2   | 1   | 2   | 1   | 2   | 2    | 1    | 2    | 2    | 1    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

## **CTUC605 - Mini Project**

### **(400 Marks)**

**Contact Hours: 12**

**Objective:**

The main objective of this mini project is to let the students apply the programming knowledge to a real-world situation/problem and exposed the students with specific programming skills and help in developing a working model in terms of application.

**Course Outcomes:**

|      |                                                                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO1: | Student will understand the implementation of concepts of SDLC and Software Engineering.                                                                |
| CO2: | The programming concepts they learn during their academics, it will be converted in to the actual implementations.                                      |
| CO3: | Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods. |
| CO4: | Students must understand the difference between a program and professional application/product/software.                                                |
| CO5: | Students will learn different categories of applications like Desktop application, Web applications, etc.                                               |

**Guidelines:**

Mini Project is in house project development. Every student is required to carry out Mini Project work under the supervision of a guide provided by the placement Coordinator. The guide shall monitor progress of the student continuously. A candidate is required to present the progress of the Mini Project work during the semester as per the schedule provided by the placement Coordinator.

Mini Project proposal should be prepared in consultation with project guide. It should clearly state the objectives and environment of proposed Mini Project to be undertaken. Project documentation must be with the respect to the project only. Project report should strictly follow the points suggested in format of project report. Placement coordinator will provide the format of project report. Student has to submit one copy of Mini Project to the

institute. Each Student is required to make a copy of Mini Project in CD and submit along with Mini Project report.

### Evaluation:

The project must be evaluated in two aspects:

- a. Internal (100 Marks):
  - i. Reporting to internal project guide
  - ii. Incorporation of suggestions by project guide
  - iii. Internal Project viva examination
- b. External (300 Marks):
  - i. Project Report Preparation & Evaluation
  - ii. External Project Viva Examination

| Course Code | Course Title | Teaching Scheme |        | Internal              | End Semester Examination |                     | Total |
|-------------|--------------|-----------------|--------|-----------------------|--------------------------|---------------------|-------|
|             |              | Contact Hours   | Credit | Continuous Evaluation | Report                   | Presentation & Viva |       |
|             | Mini Project | 12              | 12     | 200                   | 100                      | 100                 | 400   |

### Web References:

1. <http://techwhirl.com/writing-software-requirements-specifications/>[For effective SRS]
2. [http://www.ibm.com/developerworks/websphere/library/techarticles/0306\\_perks/perks2.html](http://www.ibm.com/developerworks/websphere/library/techarticles/0306_perks/perks2.html) [ For best practices of Software Project Development]
3. [http://www.uacg.bg/filebank/acadstaff/userfiles/publ\\_bg\\_397\\_SDP\\_activities\\_and\\_steps.pdf](http://www.uacg.bg/filebank/acadstaff/userfiles/publ_bg_397_SDP_activities_and_steps.pdf) [ Requirement analysis guidelines]
4. <http://www.cs.wustl.edu/~schmidt/PDF/design-principles4.pdf>[Software Design Principles and Guidelines]
5. <http://www.cse.hcmut.edu.vn/~hiep/KiemthuPhanmem/Tailieuthamkhao/Effective%20Software%20Testing%20-%2050%20specific%20ways%20to%20improve%20your%20testing.pdf>[ForEffective Software Testing]
6. [http://www.cs.uics.edu/~jbell/CourseNotes/OO\\_SoftwareEngineering/SE\\_Project\\_Report\\_Template .pdf](http://www.cs.uics.edu/~jbell/CourseNotes/OO_SoftwareEngineering/SE_Project_Report_Template.pdf)[ For guidelines to prepare software project report]

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 3   | 3   | 3   | 2   | 3   | 3   | 3   | 3   | 3   | 2    | 2    | 2    | 3    | 2    |
| CO2 | 3   | 3   | 3   | 2   | 3   | 3   | 3   | 3   | 3   | 2    | 2    | 2    | 3    | 2    |
| CO3 | 3   | 3   | 3   | 2   | 3   | 3   | 3   | 3   | 3   | 2    | 2    | 2    | 3    | 2    |
| CO4 | 3   | 3   | 3   | 2   | 3   | 3   | 3   | 3   | 3   | 2    | 2    | 2    | 3    | 2    |
| CO5 | 3   | 3   | 3   | 2   | 3   | 3   | 3   | 3   | 3   | 2    | 2    | 2    | 3    | 2    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation

**TEACHING SCHEME &  
DETAILED SYLLABUS**

**FOR**

**M.Sc. (IT) PROGRAMME  
(4<sup>th</sup> SEMESTER)**

**EFFECTIVE FROM  
ACADEMIC YEAR 2025-26**

**Teaching and Examination Scheme**  
**Master of Science in Information Technology (M.Sc.(IT)) Programme**  
**(Choice Based Credit System)**

**Effective from Year 2025-26**

**Semester-IV**

| Course Code                                                                                                                                                                | Course Title                | Teaching Scheme |          |       | Credit | CCE                   | SEE    |                     | Total |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------|----------|-------|--------|-----------------------|--------|---------------------|-------|
|                                                                                                                                                                            |                             | Contact Hours   |          |       |        | Continuous Evaluation | Report | Presentation & Viva |       |
|                                                                                                                                                                            |                             | Inst.           | Industry | Total |        |                       |        |                     |       |
| CTUC606                                                                                                                                                                    | Dissertation / Project Work | 2               | 28       | 30    | 30     | 400                   | 200    | 200                 | 800   |
| *CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation |                             |                 |          |       |        |                       |        |                     |       |

## **CTUC606 - Dissertation/Project Work**

### **(800 Marks)**

**Contact Hours: 30**

**Objective:**

The main objective of this Dissertation is to let the students apply the programming knowledge to a real-world situation/problem and exposed the students with specific programming skills and help in developing a working model in terms of application.

**Course Outcomes:**

|      |                                                                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO1: | Student will understand the implementation of concepts of SDLC and Software Engineering.                                                                |
| CO2: | The programming concepts they learn during their academics, it will be converted in to the actual implementations.                                      |
| CO3: | Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods. |
| CO4: | Students must understand the difference between a program and professional application/product/software.                                                |
| CO5: | Students will learn different categories of applications like Desktop application, Web applications, etc.                                               |

### Guidelines:

Dissertation is in house project development. Every student is required to carry out Dissertation work under the supervision of a guide provided by the placement Coordinator. The guide shall monitor progress of the student continuously. A candidate is required to present the progress of the Dissertation work during the semester as per the schedule provided by the placement Coordinator.

Dissertation proposal should be prepared in consultation with project guide. It should clearly state the objectives and environment of proposed Dissertation to be undertaken. Project documentation must be with the respect to the project only. Project report should strictly follow the points suggested in format of project report. Placement coordinator will provide the format of project report. Student has to submit one copy of Dissertation to the institute. Each Student is required to make a copy of Dissertation in CD and submit along with Dissertation report.

### Evaluation:

The project must be evaluated in two aspects:

- a. Internal (200 Marks):
  - i. Reporting to internal project guide
  - ii. Incorporation of suggestions by project guide
  - iii. Internal Project viva examination
- b. External (600 Marks):
  - i. Project Report Preparation & Evaluation
  - ii. External Project Viva Examination

| Course Code | Course Title           | Teaching Scheme |        | Internal              | End Semester Examination |                     | Total |
|-------------|------------------------|-----------------|--------|-----------------------|--------------------------|---------------------|-------|
|             |                        | Contact Hours   | Credit | Continuous Evaluation | Report                   | Presentation & Viva |       |
| CTU606      | Dissertation / Project | 30              | 30     | 200                   | 200                      | 400                 | 800   |

### Web References:

1. <http://techwhirl.com/writing-software-requirements-specifications/>[For effective SRS]
2. [http://www.ibm.com/developerworks/websphere/library/techarticles/0306\\_perks/perks2.html](http://www.ibm.com/developerworks/websphere/library/techarticles/0306_perks/perks2.html) [ For best practices of Software Project Development]
3. [http://www.uacg.bg/filebank/acadstaff/userfiles/publ\\_bg\\_397\\_SDP\\_activities\\_and\\_steps.pdf](http://www.uacg.bg/filebank/acadstaff/userfiles/publ_bg_397_SDP_activities_and_steps.pdf) [ Requirement analysis guidelines]
4. <http://www.cs.wustl.edu/~schmidt/PDF/design-principles4.pdf>[Software Design Principles and Guidelines]
5. <http://www.cse.hcmut.edu.vn/~hiep/KiemthuPhanmem/Tailieuthamkhao/Effective%20Software%20Testing%20-2050%20specific%20ways%20to%20improve%20your%20testing.pdf>[ForEffective Software Testing]
6. [http://www.cs.uics.edu/~jbell/CourseNotes/OO\\_SoftwareEngineering/SE\\_Project\\_Report\\_Template .pdf](http://www.cs.uics.edu/~jbell/CourseNotes/OO_SoftwareEngineering/SE_Project_Report_Template.pdf)[ For guidelines to prepare software project report]

### Course Articulation Matrix:

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|------|
| CO1 | 3   | 3   | 3   | 2   | 3   | 3   | 3   | 3   | 3   | 2    | 2    | 2    | 3    | 2    | -    |
| CO2 | 3   | 3   | 3   | 2   | 3   | 3   | 3   | 3   | 3   | 2    | 2    | 2    | 3    | 2    | -    |
| CO3 | 3   | 3   | 3   | 2   | 3   | 3   | 3   | 3   | 3   | 2    | 2    | 2    | 3    | 2    | -    |
| CO4 | 3   | 3   | 3   | 2   | 3   | 3   | 3   | 3   | 3   | 2    | 2    | 2    | 3    | 2    | -    |
| CO5 | 3   | 3   | 3   | 2   | 3   | 3   | 3   | 3   | 3   | 2    | 2    | 2    | 3    | 2    | -    |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation